

enter a retention period), so will meet the requirements.

upvoted 2 times

cheroh_tots 1 year, 3 months ago

Why is the answer not C?

upvoted 2 times

1dfed2b 1 year, 3 months ago

As I see, we are looking for the most operationally efficient solution. So it's B, but the most cost effective its - C (but it isn't a question).

upvoted 2 times

psyll0n 1 year, 4 months ago

https://youtu.be/g4WPLFXLwDE?si=nTWqqDcBe_Y6dtl3

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

Operational efficiency is always a managed service from AWS. AWS Backup is the right one in this case so B is right answer

upvoted 2 times

awsgeek75 1 year, 5 months ago

Answer is simply B as it if MOST operationally efficient. Other options are "distractors" to confuse everyone.

upvoted 2 times

viru 1 year, 5 months ago

Selected Answer: B

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/backuprestore_HowItWorksAWS.html

upvoted 1 times

Mikado211 1 year, 6 months ago

Selected Answer: B

Well a 7 years TTL on the dynamoDB records could be the simplest to answer the question, so B for the "retention policies". And since the B also propose AWS backup with a retention time at 7 years, why not.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

The key advantages of using AWS Backup are:

Fully managed backup service requiring minimal operational overhead

Built-in scheduling, retention policies, and backup monitoring

Supports point-in-time restore for DynamoDB

Automated and scalable solution

upvoted 1 times

tamefi5512 1 year, 11 months ago

Selected Answer: B

B - is the answer because its easy to setup via AWS Backup & It indicates the keyword "MOST Operational Efficient". Other answers are indicating Cost efficient

upvoted 1 times

cookieMr 1 year, 11 months ago

AWS Backup is a fully managed backup service that simplifies the process of creating and managing backups across various AWS services, including DynamoDB. It allows you to define backup schedules and retention policies to automatically take backups and retain them for the desired duration. By using AWS Backup, you can offload the operational overhead of managing backups to the service itself, ensuring that your data is protected and retained according to the specified retention period.

This solution is more efficient compared to the other options because it provides a centralized and automated backup management approach specifically designed for AWS services. It eliminates the need to manually configure and maintain backup processes, making it easier to ensure data retention compliance without significant operational effort.

upvoted 2 times

Rahul2212 1 year, 12 months ago

A

PITR is used to recover your table to any point in time in a rolling 35 day window, which is used to help customers mitigate accidental deletes or writes to their tables from bad code, malicious access, or user error. (==> A is the answer)

upvoted 1 times

Abrar2022 2 years ago

using AWS Backup cheaper than DynamoDB point-in-time recovery

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #79

Topic 1

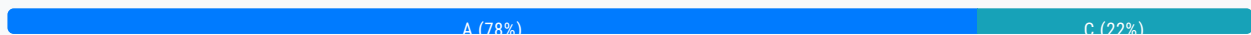
A company is planning to use an Amazon DynamoDB table for data storage. The company is concerned about cost optimization. The table will not be used on most mornings. In the evenings, the read and write traffic will often be unpredictable. When traffic spikes occur, they will happen very quickly.

What should a solutions architect recommend?

- A. Create a DynamoDB table in on-demand capacity mode. **Most Voted**
- B. Create a DynamoDB table with a global secondary index.
- C. Create a DynamoDB table with provisioned capacity and auto scaling.
- D. Create a DynamoDB table in provisioned capacity mode, and configure it as a global table.

Correct Answer: A

Community vote distribution



Comments

SimonPark **Highly Voted** 2 years, 7 months ago

Selected Answer: A

On-demand mode is a good option if any of the following are true:

- You create new tables with unknown workloads.
- You have unpredictable application traffic.
- You prefer the ease of paying for only what you use.

upvoted 52 times

123jhl0 **Highly Voted** 2 years, 7 months ago

Selected Answer: A

****A**** - On demand is the answer -

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html#HowItWorks.OnDemand>

B - not related with the unpredictable traffic

C - provisioned capacity is recommended for known patterns. Not the case here.

D - same as C

upvoted 24 times

NasosoAuxtyno 2 years, 3 months ago

Thanks. Your reference link perfectly supports the option "A". 100% correct
upvoted 3 times

patdevops Most Recent 3 months ago

Selected Answer: A

The correct answer is A, keyword "unpredictable workloads"
upvoted 1 times

AshishDhole 5 months ago

Selected Answer: A

Unknown Load/traffic = On demand mode
upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: A

On-demand mode of DynamoDB is the most suitable for unpredictable workloads.
upvoted 1 times

JA2018 6 months, 2 weeks ago

Selected Answer: C

after browsing through all the responses, I decided to vote for C for the following 2 reasons:

- Unpredictable traffic: Since the evening traffic is often unpredictable with quick spikes, using provisioned capacity with auto scaling allows the table to automatically adjust to meet the fluctuating demand, preventing potential throttling issues during high traffic periods.
 - Cost optimization: By utilizing auto scaling, the company only pays for the capacity it actually uses, which is ideal for a scenario where the table experiences low usage during mornings and high, unpredictable usage in the evenings.
- upvoted 2 times

JA2018 6 months, 2 weeks ago

For Option A, come to think about it, while it is true that on-demand capacity can handle spikes, it may not be the most cost-effective option as you still have to pay for the capacity used even during low traffic periods.
upvoted 1 times

PaulGa 9 months ago

Selected Answer: A

Ans A - handles unpredictable workloads/traffic; pay on demand
C - provisioned capacity for known data I/O - not the case here
upvoted 2 times

SaurabhTiwari1 9 months, 2 weeks ago

Selected Answer: A

Unpredictable = On-demand
upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

cost is concern so CD are not right as provisioning is costly
B is irrelevant
A on-demand is correct as it will scale according to the usage pattern which is from low to very abrupt high
upvoted 3 times

MiniYang 1 year, 6 months ago

Selected Answer: C

Choosing the On-Demand Capacity model (Option A) may cause performance issues during peak periods because it relies on DynamoDB to automatically adjust throughput based on actual usage, which may not be able to cope with sudden traffic increases in time.

Choosing a DynamoDB table with a global secondary index (option B) is independent of the capacity model and does not directly solve the problem of peak traffic.

Choosing to build DynamoDB tables in provisioned capacity mode and configure them as global tables (option D) may increase costs in some cases without necessarily providing the flexibility to accommodate unpredictable peak traffic.

upvoted 1 times

MiniYang 1 year, 6 months ago

sorry I want to change my answer to A . Because the point is the " cost "

upvoted 7 times

BartoszGolebiowski24 1 year, 7 months ago

Selected Answer: A

DynamoDB autoscaling takes 2 minutes to increase capacity. We need to handle it immediately.

"Application Auto Scaling automatically scales the provisioned capacity only when the consumed capacity is higher than target utilization for two consecutive minutes".

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TroubleshootingThrottling.html>

upvoted 4 times

xdkonorek2 1 year, 7 months ago

This is important, thank you for the link. A definitely

upvoted 2 times

Wayne23Fang 1 year, 8 months ago

Selected Answer: A

The costly part of (C) is you need to pay for what you order not what you have used for (A) On-Demand: A reserved capacity purchase is an agreement to pay for a minimum amount of provisioned throughput capacity, for the duration of the term of the agreement, in exchange for discounted pricing. If you use less than your reserved capacity, you will still be charged each month for that minimum amount of provisioned throughput capacity.

upvoted 2 times

clark777 1 year, 8 months ago

<https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html>

With on-demand capacity mode, DynamoDB charges you for the data reads and writes your application performs on your tables. You do not need to specify how much read and write throughput you expect your application to perform because DynamoDB instantly accommodates your workloads as they ramp up or down.

With provisioned capacity mode, you specify the number of reads and writes per second that you expect your application to require, and you are billed based on that. Furthermore if you can forecast your capacity requirements you can also reserve a portion of DynamoDB provisioned capacity and optimize your costs even further.

upvoted 3 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: A

With on-demand capacity mode, DynamoDB instantly accommodates your workloads as they ramp up or down.

upvoted 2 times

ontheyun 1 year, 11 months ago

on-demand capacity : unpredictable application traffic

provisioned capacity : predictable application traffic, run applications whose traffic is consistent, and ramps up or down gradually.

<https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html>

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: C

By choosing provisioned capacity, you can allocate a specific amount of read and write capacity units based on your expected usage during peak times. This helps in cost optimization as you only pay for the provisioned capacity, which can be adjusted according to your anticipated traffic.

Enabling auto scaling allows DynamoDB to automatically adjust the provisioned capacity up or down based on the actual

Enabling auto scaling allows DynamoDB to automatically adjust the provisioned capacity up or down based on the actual usage. This is beneficial in handling quick traffic spikes without manual intervention and ensuring that the required capacity is available to handle increased load efficiently. Auto scaling helps to optimize costs by dynamically adjusting the capacity to match the demand, avoiding overprovisioning during periods of low usage.

A. Creating a DynamoDB table in on-demand capacity mode, may not be the most cost-effective solution in this scenario. On-demand capacity mode charges you based on the actual usage of read and write requests, which can be beneficial for sporadic or unpredictable workloads. However, it may not be the optimal choice if the table is not used on most mornings.

upvoted 10 times

beginnercloud 2 years ago

Selected Answer: A

Correct answer is A

- You create new tables with unknown workloads. - You have unpredictable application traffic. - You prefer the ease of paying for only what you use.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #80

Topic 1

A company recently signed a contract with an AWS Managed Service Provider (MSP) Partner for help with an application migration initiative. A solutions architect needs to share an Amazon Machine Image (AMI) from an existing AWS account with the MSP Partner's AWS account. The AMI is backed by Amazon Elastic Block Store (Amazon EBS) and uses an AWS Key Management Service (AWS KMS) customer managed key to encrypt EBS volume snapshots.

What is the MOST secure way for the solutions architect to share the AMI with the MSP Partner's AWS account?

A. Make the encrypted AMI and snapshots publicly available. Modify the key policy to allow the MSP Partner's AWS account to use the key.

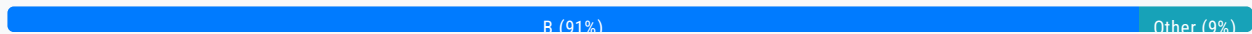
B. Modify the launchPermission property of the AMI. Share the AMI with the MSP Partner's AWS account only. Modify the key policy to allow the MSP Partner's AWS account to use the key. **Most Voted**

C. Modify the launchPermission property of the AMI. Share the AMI with the MSP Partner's AWS account only. Modify the key policy to trust a new KMS key that is owned by the MSP Partner for encryption.

D. Export the AMI from the source account to an Amazon S3 bucket in the MSP Partner's AWS account, Encrypt the S3 bucket with a new KMS key that is owned by the MSP Partner. Copy and launch the AMI in the MSP Partner's AWS account.

Correct Answer: B

Community vote distribution

**Comments**

Sauran **Highly Voted** 2 years, 7 months ago

Selected Answer: B

Share the existing KMS key with the MSP external account because it has already been used to encrypt the AMI snapshot.

<https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html>

upvoted 20 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: B

CORRECT

B. Modify the launchPermission property of the AMI.

The most secure way for the solutions architect to share the AMI with the MSP Partner's AWS account would be to modify the launchPermission property of the AMI and share it with the MSP Partner's AWS account only. The key policy should also be modified to allow the MSP Partner's AWS account to use the key. This ensures that the AMI is only shared with the MSP Partner and is encrypted with a key that they are authorized to use.

upvoted 15 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it.

Option C, modifying the key policy to trust a new KMS key owned by the MSP Partner, is also not a secure option as it would involve sharing the key with the MSP Partner, which could potentially compromise the security of the data encrypted with the key.

Option D, exporting the AMI to an S3 bucket in the MSP Partner's AWS account and encrypting the S3 bucket with a new KMS key owned by the MSP Partner, is also not the most secure option as it involves sharing the AMI and a new key with the MSP Partner, which could potentially compromise the security of the data.

upvoted 17 times

Gizmo2022 6 months, 2 weeks ago

Thank u so much for explaining this Buruguduy

upvoted 3 times

jerry! Most Recent 2 months, 3 weeks ago

Selected Answer: C

Can anyone explain to me? I am not familiar with the terminology

I will go for C, because B mention sharing the key to partner, which might expose other resource which use the same key for encryption, or is this scenario not possible to happen (i.e. key will not be reused across resources)?

Thanks

upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: B

I got this wrong! I thought C is right but it turns out that is not secure.

In this question, or any question with encryption, one must consider which of the available options is the most secure way, and it turns out that in this case it is B.

upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: B

I got this wrong, select D initially, but after read the comments, I agree B is the right solution.

1. <https://docs.aws.amazon.com/toolkit-for-visual-studio/latest/user-guide/tkv-set-ami-launch-perms.html>

2. <https://docs.aws.amazon.com/toolkit-for-visual-studio/latest/user-guide/tkv-set-ami-launch-perms.html>

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: B

Updating the key policy is the most secure way to get access here.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: B

Ans B - keep the control simple by only allowing MSP Partner access to the key

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

AD are unsecure.

I was confused between B and C but read the article (link below). You have to allow the other account to use your key somehow otherwise they won't be able to use your AMI. C just allows a trust relationship with MSP's KMS, it won't give them access to your key.

your key.

<https://aws.amazon.com/blogs/security/how-to-share-encrypted-amis-across-accounts-to-launch-encrypted-ec2-instances/>

upvoted 4 times

xdkonorek2 1 year, 7 months ago

Selected Answer: B

when you export AMI to s3 bucket it remains encrypted, so partner couldn't launch ec2 instance

upvoted 3 times

Ruffyit 1 year, 7 months ago

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/sharingamis-explicit.html>

upvoted 2 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: B

Share the AMI and Key with the MSP Partner's AWS account only

upvoted 2 times

tamefi5512 1 year, 11 months ago

Selected Answer: B

B - is the Answer

<https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html>

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: B

By modifying the launchPermission property of the AMI and sharing it with the MSP Partner's account only, the solutions architect restricts access to the AMI and ensures that it is not publicly available.

Additionally, modifying the key policy to allow the MSP Partner's account to use KMS customer managed key used for encrypting the EBS snapshots ensures that the MSP Partner has the necessary permissions to access and use the key for decryption.

upvoted 3 times

Abrar2022 2 years ago

CORRECTION to my last comment Option B is correct not A.

Explanation why..

Making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it. Best practice would be to share the AMI with the MSP Partner's AWS account then Modify launchPermission property of the AMI. This ensures that the AMI is shared only with the MSP Partner and is encrypted with a key that they are authorised to use.

upvoted 3 times

Abrar2022 2 years ago

Selected Answer: A

Option A, making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it. Best practice would be to share the AMI with the MSP Partner's AWS account then Modify launchPermission property of the AMI. This ensures that the AMI is shared only with the MSP Partner and is encrypted with a key that they are authorised to use.

upvoted 1 times

draum010 2 years, 2 months ago

Selected Answer: D

Option D

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #81

Topic 1

A solutions architect is designing the cloud architecture for a new application being deployed on AWS. The process should run in parallel while adding and removing application nodes as needed based on the number of jobs to be processed. The processor application is stateless. The solutions architect must ensure that the application is loosely coupled and the job items are durably stored.

Which design should the solutions architect use?

- A. Create an Amazon SNS topic to send the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch configuration that uses the AMI. Create an Auto Scaling group using the launch configuration. Set the scaling policy for the Auto Scaling group to add and remove nodes based on CPU usage.
- B. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch configuration that uses the AMI. Create an Auto Scaling group using the launch configuration. Set the scaling policy for the Auto Scaling group to add and remove nodes based on network usage.
- C. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of items in the SQS queue. **Most Voted**
- D. Create an Amazon SNS topic to send the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of messages published to the SNS topic.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Marge_Simpson **Highly Voted** 2 years, 6 months ago

Selected Answer: C

decoupled = SQS
launch template = AMI

Launch template = AMI
Launch configuration = EC2
upvoted 55 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: C

This design follows the best practices for loosely coupled and scalable architecture. By using SQS, the jobs are durably stored in the queue, ensuring they are not lost. The processor application is stateless, which aligns with the design requirement. The AMI allows for consistent deployment of the application. The launch template and ASG facilitate the dynamic scaling of the application based on the number of items in the SQS, ensuring parallel processing of jobs.

Options A and D suggest using SNS, which is a publish/subscribe messaging service and may not provide the durability required for job storage.

Option B suggests using network usage as a scaling metric, which may not be directly related to the number of jobs to be processed. The number of items in the SQS provides a more accurate metric for scaling based on the workload.

upvoted 9 times

PaulGa Most Recent 9 months ago

Selected Answer: C

Ans C - decoupled and durable (SQS), auto-scaling based on number of messages.

upvoted 3 times

setout4saa 1 year ago

Selected Answer: C

C for sure

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

SNS is not reliable in case of processing failure which is why none of the SNS options are useful. SQS (not in FIFO mode) allows parallel message processing but reliability/durability of messages is guaranteed. AMI/EC2 scaling is good choice and scaling parameter should be number of messages. Hence "C" is the correct answer

upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: C

"Based on the number of jobs to be processed" -> that alone rules out ABD because only C is based on queue length. (D is based on "number of messages published to the queue", not number of messages currently in queue.)

"Job items are durably stored" also speaks for SQS over SNS.

upvoted 3 times

clumsyninja4life 1 year, 5 months ago

so many words...

upvoted 2 times

awsgeek75 1 year, 5 months ago

... yet only one answer is correct

upvoted 3 times

slimen 1 year, 7 months ago

Selected Answer: C

from my perspective, I didn't go for D even though it provides decoupled architecture is because in the question they said "parallel processing"

SNS sends the same message to all the subscribers, but in this case we don't want all the nodes to process the same message instead we want them to process as much jobs as possible in a parallel fashion.

SQS in this case is more suitable because each job will get a message and process it and the next message will be taken by another job and so on..

upvoted 4 times

dashku 1 year, 8 months ago

uarekw 1 year, 9 months ago

<https://aws.amazon.com/about-aws/whats-new/2021/03/aws-certificate-manager-provides-certificate-expiry-monitoring-through-amazon-cloudwatch/>

upvoted 3 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: C

Loosely coupled = Amazon SQS queue

New application being deployed = deploy on Amazon Machine Image

Adding and removing application nodes as needed based on the number of jobs to be processed = Auto Scaling group with launch template

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

The recommended design is to use an SQS queue to store jobs (option C):

SQS provides a durable and decoupled queue to store job items

An Auto Scaling group with scaling policies based on SQS queue length will add/remove nodes as needed

Launch templates provide flexibility to update AMIs

The key points:

SQS enables loose coupling and stores jobs durably

Auto Scaling provides parallel processing

Scaling based on queue length manages nodes effectively

upvoted 3 times

Bmarodi 2 years ago

Selected Answer: C

C for sure

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: C

CORRECT

The correct design is Option C. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of items in the SQS queue.

This design satisfies the requirements of the application by using Amazon Simple Queue Service (SQS) as durable storage for the job items and Amazon Elastic Compute Cloud (EC2) Auto Scaling to add and remove nodes based on the number of items in the queue. The processor application can be run in parallel on multiple nodes, and the use of launch templates allows for flexibility in the configuration of the EC2 instances.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG

Option A is incorrect because it uses Amazon Simple Notification Service (SNS) instead of SQS, which is not a durable storage solution.

Option B is incorrect because it uses CPU usage as the scaling trigger instead of the number of items in the queue.

Option D is incorrect for the same reasons as option A.

upvoted 7 times

graveend 1 year, 10 months ago

SNS provides durable storage of all messages that it receives.

Ref:

<https://aws.amazon.com/sns/faqs/#:~:text=SNS%20provides%20durable%20storage%20of%20all%20messages%20that%20it%20receives.>

Why use SQS instead of SNS? In the question it says parallel execution of processes. SNS has that ability.

why use SQS instead of SNS: in the question it says parallel execution of processes. SNS has that ability.
upvoted 1 times

cyber_bedouin 1 year, 7 months ago

SQS satisfies the decoupling requirement
upvoted 1 times

awsgeek75 1 year, 5 months ago

SNS is not a durable storage. SQS stores the messages until they are process. SNS just notifies the subscribers and doesn't care if the notification is processed or not. It's kind of "fire and forget"
upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: C

SQS with EC2 autoscaling policy based number of messages in the queue.
upvoted 1 times

Uhrien 2 years, 6 months ago

Selected Answer: C

C is correct
upvoted 2 times

kelljons 2 years, 6 months ago

what about the word "coupled"
upvoted 1 times

kewl 2 years, 6 months ago

Selected Answer: C

AWS strongly recommends that you do not use launch configurations hence answer is C
https://docs.amazonaws.cn/en_us/autoscaling/ec2/userguide/launch-configurations.html
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #82

Topic 1

A company hosts its web applications in the AWS Cloud. The company configures Elastic Load Balancers to use certificates that are imported into AWS Certificate Manager (ACM). The company's security team must be notified 30 days before the expiration of each certificate.

What should a solutions architect recommend to meet this requirement?

- A. Add a rule in ACM to publish a custom message to an Amazon Simple Notification Service (Amazon SNS) topic every day, beginning 30 days before any certificate will expire.
- B. Create an AWS Config rule that checks for certificates that will expire within 30 days. Configure Amazon EventBridge (Amazon CloudWatch Events) to invoke a custom alert by way of Amazon Simple Notification Service (Amazon SNS) when AWS Config reports a noncompliant resource.
- C. Use AWS Trusted Advisor to check for certificates that will expire within 30 days. Create an Amazon CloudWatch alarm that is based on Trusted Advisor metrics for check status changes. Configure the alarm to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS).
- D. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to detect any certificates that will expire within 30 days. Configure the rule to invoke an AWS Lambda function. Configure the Lambda function to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS). **Most Voted**

Correct Answer: D

Community vote distribution



Comments

LeGloupier **Highly Voted** 2 years, 7 months ago

B
 AWS Config has a managed rule
 named acm-certificate-expiration-check
 to check for expiring certificates
 (configurable number of days)
 upvoted 92 times

Mia2009687 1 year, 11 months ago

B costs more than D

To get a notification that your certificate is about to expire, use one of the following methods:

Use the ACM API in Amazon EventBridge to configure the ACM Certificate Approaching Expiration event.
Create a custom EventBridge rule to receive email notifications when certificates are nearing the expiration date.
Use AWS Config to check for certificates that are nearing the expiration date.
If you use AWS Config for this resolution, then be aware of the following:

Before you set up the AWS Config rule, create the Amazon Simple Notification Service (Amazon SNS) topic and EventBridge rule. This makes sure that all non-compliant certificates invoke a notification before the expiration date.
Activating AWS Config incurs an additional cost based on usage. For more information, see AWS Config pricing.
<https://repost.aws/knowledge-center/acm-certificate-expiration>
upvoted 5 times

pentium75 1 year, 5 months ago

Nobody asked for cost optimization.
upvoted 12 times

824c449 1 year, 1 month ago

It does not have a built-in rule for checking the expiration of ACM certificates directly.
upvoted 2 times

LeGloupier 2 years, 7 months ago

<https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>
upvoted 15 times

ChrisG1454 2 years, 3 months ago

Answer B and answer D are possible according to this article.
So, need to read B & D carefully to determine the most suitable answer.

Reference: <https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>
upvoted 9 times

Bayebrymo 1 year, 1 month ago

But from the link shared. It doesn't state that you should invoke lambda function as stated in option D. Option B is explicitly stated in the article as also worded in option B. So I think B should be the answer. My thought though. I stand to be corrected.
upvoted 8 times

TTaws 1 year, 11 months ago

It's B, simply because in option D - event bridge cannot "detect" anything.
upvoted 9 times

RupeC 1 year, 10 months ago

My understanding is that the ACM sends a Cert Expiration event to EventBridge. Thus EB. does not need to detect anything.
upvoted 3 times

pentium75 1 year, 5 months ago

"ACM sends a Cert Expiration event to EventBridge" yes, but 45 (not 30) days before expiration.
upvoted 2 times

mrkmtai 1 year, 2 months ago

This can be configured to whatever you need in the Days to expiry box
upvoted 2 times

darekw 1 year, 9 months ago

AWS Certificate Manager (ACM) now publishes certificate metrics and events through Amazon CloudWatch and Amazon EventBridge.

<https://aws.amazon.com/about-aws/whats-new/2021/03/aws-certificate-manager-provides-certificate-expiry-monitoring->

through-amazon-cloudwatch/
upvoted 4 times

ManoAni Highly Voted 2 years, 7 months ago

Selected Answer: B

<https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>
upvoted 18 times

sagarvh Most Recent 3 weeks, 6 days ago

Selected Answer: D

Answer is D
upvoted 1 times

network_enthusiast 2 months ago

Selected Answer: D

Amazon Event Bridge can detect certificate expiry event for certificate managed by ACM.
upvoted 1 times

dattateja8 2 months, 2 weeks ago

Selected Answer: B

B
AWS Config has a managed rule
named acm-certificate-expiration-check
to check for expiring certificates
(configurable number of days)
upvoted 1 times

dattateja8 2 months, 2 weeks ago

Selected Answer: D

AWS Config does not have in-built feature to check certificate expiry and it has to use out of the box managed rule to check like AWS Lambda. In Answer B it is not mentioned to use AWS Lambda. So the best option here is D.
upvoted 1 times

gigilla 3 months, 1 week ago

Selected Answer: B

B
AWS Config has a managed rule
named acm-certificate-expiration-check
to check for expiring certificates
(configurable number of days)
upvoted 1 times

kg508 3 months, 2 weeks ago

Selected Answer: D

B) EventBridge, AWS Config, AWS Lambda, Amazon SNS, Amazon SES, Amazon SES, Amazon SES, Amazon SES
eventbridge, AWS Config, AWS Lambda, Amazon SNS, Amazon SES, Amazon SES, Amazon SES, Amazon SES
upvoted 1 times

iamroyalty_k 4 months ago

Selected Answer: D

Why Option D is Correct?

- EventBridge can trigger an alert when an ACM certificate is about to expire.
- Lambda can process the event and send a notification through Amazon SNS to notify the security team.
- SNS ensures the notifications are delivered via email, SMS, or other means based on the team's preference.
- This is an automated, scalable, and AWS-native solution.

Why Not the Other Options?

- A - □ ACM does not support custom rules or direct SNS integration for expiring certificates.
- B - □ AWS Config does not have a built-in rule to check ACM certificate expiration. Creating a custom rule would add unnecessary complexity.
- C - □ EventBridge can trigger an alert when an ACM certificate is about to expire, but it cannot send a notification through Amazon SNS to notify the security team.

C - LI Trusted Advisor does not check ACM certificates for expiration. It only monitors IAM certificate expiration.
upvoted 2 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: B

B or D both will work. Per the documentation:

To get a notification that your certificate is about to expire, use one of the following methods:

Use the ACM API in Amazon EventBridge to configure the ACM Certificate Approaching Expiration event.
Create a custom EventBridge rule to receive email notifications when certificates are near the expiration date.
Use AWS Config to check for certificates that are near the expiration date.
Create an Amazon CloudWatch alarm based on a static threshold when certificates are near the expiration date.
upvoted 1 times

itsmeDiyan 4 months, 2 weeks ago

Selected Answer: D

Use EventBridge, Lambda, and SNS
upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: C

Based on this article it is C if the notification is within 45 days, otherwise use B
<https://repost.aws/knowledge-center/acm-certificate-expiration>
upvoted 1 times

iagotb 4 months, 2 weeks ago

Selected Answer: B

Correct answer is B.

Based on <https://repost.aws/knowledge-center/acm-certificate-expiration>

D is not valid because by default AWS Config will send ACM Certificate Approaching Expiration event at 45 days. Question asks for 30 days so then we will need to:

After you create the rule, you can change the timing of the expiration notification. In the ACM API's PutAccountConfiguration action, enter a value between 1-45 for DaysBeforeExpiry.

As D answer doesn't say that is not valid.

In the other hand, answer B is completely valid.
upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: D

Based on this article, D is a better solution. reasons:
- AWS Config has charges
- AWS config can only be configured for the day, so D will be more real time.
upvoted 1 times

d8 4 months, 3 weeks ago

Selected Answer: D

Option D:
Fully automated and customizable solution.
Real-time monitoring and notifications.
Scalable and cost-efficient.
upvoted 1 times

AshishDhole 4 months, 4 weeks ago

Selected Answer: B

Answer Should be : B
Create a custom EventBridge rule to receive email notifications when certificates are near the expiration date.

Use AWS Config to check for certificates that are near the expiration date.

Create an Amazon CloudWatch alarm based on a static threshold when certificates are near the expiration date
upvoted 1 times

ricktechie66 5 months, 1 week ago

Selected Answer: D

D is incorrect because EventBridge alone cannot directly detect certificate expiration - it needs to work with another service (like AWS Config) to get this information

upvoted 1 times

rindatta 5 months, 1 week ago

No, EventBridge can do that alone with configuring the "ACM Certificate Approaching Expiration" event in EventBridge. Ceck the below link:

<https://repost.aws/knowledge-center/acm-certificate-expiration>

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #83

Topic 1

A company's dynamic website is hosted using on-premises servers in the United States. The company is launching its product in Europe, and it wants to optimize site loading times for new European users. The site's backend must remain in the United States. The product is being launched in a few days, and an immediate solution is needed.

What should the solutions architect recommend?

- A. Launch an Amazon EC2 instance in us-east-1 and migrate the site to it.
- B. Move the website to Amazon S3. Use Cross-Region Replication between Regions.
- C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers. **Most Voted**
- D. Use an Amazon Route 53 geoproximity routing policy pointing to on-premises servers.

Correct Answer: C*Community vote distribution*

C (100%)

Comments**Buruguduystunstugudunstuy** **Highly Voted** 2 years, 5 months ago**Selected Answer: C**

CORRECT

C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers.

Amazon CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as HTML, CSS, JavaScript, images, and videos. By using CloudFront, the company can distribute the content of their website from edge locations that are closer to the users in Europe, reducing the loading times for these users.

To use CloudFront, the company can set up a custom origin pointing to their on-premises servers in the United States. CloudFront will then cache the content of the website at edge locations around the world and serve the content to users from the location that is closest to them. This will allow the company to optimize the loading times for their European users without having to move the backend of the website to a different region.

upvoted 40 times

Euowellima 1 year, 9 months ago

excelente explicação

upvoted 3 times

TarinKinkamai 2 years, 3 months ago

tanaykumar 2 years, 5 months ago

good explanation..thanks
upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG

Option A (launch an Amazon EC2 instance in us-east-1 and migrate the site to it) would not address the issue of optimizing loading times for European users.

Option B (move the website to Amazon S3 and use Cross-Region Replication between Regions) would not be an immediate solution as it would require time to set up and migrate the website.

Option D (use an Amazon Route 53 geoproximity routing policy pointing to on-premises servers) would not be suitable because it would not improve the loading times for users in Europe.

upvoted 19 times

SVDK 1 year, 5 months ago

S3 doesn't support dynamic website hosting. Therefore, can be ruled out.
upvoted 5 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: C

C. This solution leverages the global network of CloudFront edge locations to cache and serve the website's static content from the edge locations closest to the European users.

A. Hosting the website in a single region would still result in increased latency for European users accessing the site.

B. Moving the website to S3 and implementing Cross-Region Replication would distribute the website's content across multiple regions, including Europe. S3 is primarily used for static content hosting, and it does not provide server-side processing capabilities necessary for dynamic website functionality.

D. Using a geoproximity routing policy in Route 53 would allow you to direct traffic to the on-premises servers based on the geographic location of the users. However, this option does not optimize site loading times for European users as it still requires them to access the website from the on-premises servers in the United States. It does not leverage the benefits of content caching and edge locations for improved performance.

upvoted 5 times

satyaamm **Most Recent** 5 months, 1 week ago

Selected Answer: C

CloudFront is the most suitable for reducing latency here.
upvoted 2 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: C

C for sure
upvoted 1 times

PaulGa 9 months ago

Selected Answer: C

Ans C...

I was going for D, then I saw (CookieMr, 1yr 2mth ago) "Route 53 geoproximity routing policy in does allow you to direct traffic to the on-premises servers based on the geographic location of the users - BUT does not optimised site loading times for European users as it still requires them to access the website from the on-prem servers in the US."

upvoted 2 times

Hkayne 1 year, 1 month ago

Selected Answer: C

C is the appropriate answer
upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

Immediate solution is C

A: Requires site migration so won't be immediate

B: Dynamic site cannot work from S3

D: Geoproximity routing policy finds a server close to the user so this makes no sense

C: Not ideal but best option given the "immediate" requirement as CloudFront is a CDN so it will cache whatever is possible as close to the user giving best performance in this case (i.e. when compared to other options)

upvoted 3 times

Ruffyt 1 year, 7 months ago

C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

The key reasons are:

CloudFront can cache static content close to European users using edge locations, improving site performance.

The custom origin feature allows seamlessly integrating the CloudFront CDN with existing on-premises servers.

No changes are needed to the site backend or servers. CloudFront just acts as a globally distributed cache.

This can be set up very quickly, meeting the launch deadline.

Other options like migrating to EC2 or S3 would require more time and changes. CloudFront is an easier lift.

Route 53 geoproximity routing alone would not improve performance much without a CDN.

upvoted 3 times

Bmarodi 2 years ago

Selected Answer: C

C is best solution.

upvoted 1 times

gustavtd 2 years, 5 months ago

Selected Answer: C

Within few days you can not do more than using CloudFront

upvoted 3 times

career360guru 2 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

kajal1206 2 years, 6 months ago

Selected Answer: C

C is correct answer

upvoted 1 times

koreanmonkey 2 years, 6 months ago

Selected Answer: C

CloudFront = CDN Service

upvoted 3 times

Liliwood 2 years, 6 months ago

C.

S3 Cross region Replication minimize latency but also copies objects across Amazon S3 buckets in different AWS Regions(data has to remain in origin thou) so B wrong.

Route 53 geo, does not help reducing the latency.

upvoted 3 times

Wpcorgan 2 years, 6 months ago

C is correct

upvoted 1 times

Hunkie 2 years, 7 months ago

Same question with detailed explanation

<https://www.examtopics.com/discussions/amazon/view/27898-exam-aws-certified-solutions-architect-associate-saa-c02/>
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #84

Topic 1

A company wants to reduce the cost of its existing three-tier web architecture. The web, application, and database servers are running on Amazon EC2 instances for the development, test, and production environments. The EC2 instances average 30% CPU utilization during peak hours and 10% CPU utilization during non-peak hours. The production EC2 instances run 24 hours a day. The development and test EC2 instances run for at least 8 hours each day. The company plans to implement automation to stop the development and test EC2 instances when they are not in use. Which EC2 instance purchasing solution will meet the company's requirements MOST cost-effectively?

- A. Use Spot Instances for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
- B. Use Reserved Instances for the production EC2 instances. Use On-Demand Instances for the development and test EC2 instances. **Most Voted**
- C. Use Spot blocks for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
- D. Use On-Demand Instances for the production EC2 instances. Use Spot blocks for the development and test EC2 instances.

Correct Answer: B

Community vote distribution



Comments

ArielSchivo **Highly Voted** 2 years, 7 months ago

Selected Answer: B

Spot blocks are not longer available, and you can't use spot instances on Prod machines 24x7, so option B should be valid.
upvoted 20 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: B

Option B, would indeed be the most cost-effective solution. Reserved Instances provide cost savings for instances that run consistently, such as the production environment in this case, while On-Demand Instances offer flexibility and are suitable for instances with variable usage patterns like the development and test environments. This combination ensures cost optimization based on the specific requirements and usage patterns described in the question.

upvoted 9 times

devmon 1 year, 9 months ago

In addition to this, we can set up an automated process to start and stop the EC2 instances in the test and dev environment
upvoted 3 times

satyaamm Most Recent 5 months, 1 week ago

Selected Answer: B

Using Reserved Instances is suitable for predictable workloads while using On-demand is suitable for dev and test instances as we want to reduce their costs by paying for only resources that we use.

upvoted 1 times

EllenLiu 6 months ago

Selected Answer: B

production - reserved
test/dev - on-demand

upvoted 1 times

PaulGa 9 months ago

Selected Answer: B

Ans B -

A: Spot instances are not ideal for production due to possibility of not running

C,D: Spot blocks discontinued.

That leaves option B

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: B

B is the right answer

upvoted 1 times

KTEgghead 11 months, 3 weeks ago

Either a trick question, or an old question. The answer is B; spot blocks were discontinued by Amazon, July 2021.

upvoted 3 times

TheTeaBoy 7 months ago

Indeed, and anyway Spot blocks only run for a finite duration (1-6 hours), the question says these development and test instances need to run for at least 8, so that makes B the correct answer (even if Spot Blocks were still supported).

<https://aws.amazon.com/blogs/aws/new-ec2-spot-blocks-for-defined-duration-workloads/>

upvoted 2 times

MehulKapadia 1 year, 2 months ago

Selected Answer: B

Option A,C: Production workload must not run on spot instances.

Option D: No savings(compared to reserved Instances) for prod as its on-demand, Spot Blocks are no longer available.

So correct answer is B

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

Isn't this simple or am I thinking wrong?

"The production EC2 instances run 24 hours a day"

AC are not going to give the 24 hour usage as spot is for intermittent pattern.

D is just normal cost without any discounts for production

B use "reserved" instances so there is an option for discount in billing. On-demand for dev/test is ok as their usage patter doesn't really fall in reserved or spot usage discounts

upvoted 3 times

MrPCarrot 1 year, 6 months ago

B = Reserved for Prod and On Demand for Dev
upvoted 5 times

Bmarodi 2 years ago

Selected Answer: B

B meets the requirements, and most cost-effective.
upvoted 2 times

ChanghyeonYoon 2 years, 1 month ago

Selected Answer: B

Spot instances are not suitable for production due to the possibility of not running.
upvoted 3 times

alexiscloud 2 years, 2 months ago

Answer B:
Spot block are not longer available and you can't use spot instace on production
upvoted 2 times

Nandan747 2 years, 5 months ago

Selected Answer: B

Well, AWS has DISCONTINUED the Spot-Block option. so that rules out the two options that use spot-block. Wait, this question must be from SAA-C02 or even 01. STALE QUESTION. I don't think this will feature in SAA-C03. Anyhow, the most cost-effective solution would be Option "b"
upvoted 7 times

Wajif 2 years, 5 months ago

Selected Answer: B

Choosing B as spot blocks (Spot instances with a finite duration) are no longer offered since July 2021
upvoted 2 times

sparky231 2 years ago

https://aws.amazon.com/ec2/spot/?cards.sort-by=item.additionalFields.startDateTime&cards.sort-order=asc&trk=8e336330-37e5-41e0-8438-bc1c75320d09&sc_channel=ps&ef_id=CjwKCAjw67ajBhAVEiwA2g_jECgIX_lcbqawbH-wVx2Y_EozBm8xv3g3Ci1eps0V49XcZRyfuy9xPhoCOkcQAvD_BwE:G:s&s_kwcid=AL!4422!3!517520538467!p!!g!!aws%20ec2%20spot!12831094520!122300635918
upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: A

The most cost-effective solution for the company's requirements would be to use Spot Instances for the development and test EC2 instances and Reserved Instances for the production EC2 instances.

Spot Instances are a cost-effective choice for non-critical, flexible workloads that can be interrupted. Since the development and test EC2 instances are only needed for at least 8 hours per day and can be stopped when not in use, they would be a good fit for Spot Instances.
upvoted 2 times

vincent2023 5 months ago

Spot instances can't be used for production environment as they are not long lived in nature.
upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Reserved Instances are a good fit for production EC2 instances that need to run 24 hours a day, as they offer a significant discount compared to On-Demand Instances in exchange for a one-time payment and a commitment to use the instances for a certain period of time.

Option A is the correct answer because it meets the company's requirements for cost-effectively running the development and test EC2 instances and the production EC2 instances.
upvoted 2 times

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option B is not the most cost-effective solution because it suggests using On-Demand Instances for the development and test EC2 instances, which would be more expensive than using Spot Instances. On-Demand Instances are a good choice for workloads that require a guaranteed capacity and can't be interrupted, but they are more expensive than Spot Instances.

Option C is not the correct solution because Spot blocks are a variant of Spot Instances that offer a guaranteed capacity and duration, but they are not available for all instance types and are not necessarily the most cost-effective option in all cases. In this case, it would be more cost-effective to use Spot Instances for the development and test EC2 instances, as they can be interrupted when not in use.

upvoted 1 times

WherecanIstart 2 years, 3 months ago

Can't use Spot instances for Production environment that needs to run 24/7. That should tell you that Production instances can't have a downtime. Spot instances are used when an application or service can allow disruption and 24/7 production environment won't allow that.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option D is not the correct solution because it suggests using On-Demand Instances for the production EC2 instances, which would be more expensive than using Reserved Instances. On-Demand Instances are a good choice for workloads that require a guaranteed capacity and can't be interrupted, but they are more expensive than Reserved Instances in the long run. Using Reserved Instances for the production EC2 instances would offer a significant discount compared to On-Demand Instances in exchange for a one-time payment and a commitment to use the instances for a certain period of time.

upvoted 1 times

AWSSURI 10 months, 3 weeks ago

Reserved instances are a better choice when compared to spot instances because spot instance cannot be used for 24/7 running instances as stated in this scenario.

upvoted 2 times

PassNow1234 2 years, 5 months ago

The production EC2 instances run 24 hours a day.

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #85

Topic 1

A company has a production web application in which users upload documents through a web interface or a mobile app. According to a new regulatory requirement, new documents cannot be modified or deleted after they are stored. What should a solutions architect do to meet this requirement?

- A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled. **Most Voted**
- B. Store the uploaded documents in an Amazon S3 bucket. Configure an S3 Lifecycle policy to archive the documents periodically.
- C. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning enabled. Configure an ACL to restrict all access to read-only.
- D. Store the uploaded documents on an Amazon Elastic File System (Amazon EFS) volume. Access the data by mounting the volume in read-only mode.

Correct Answer: A*Community vote distribution*

A (100%)

Comments**123jh10** **Highly Voted** 2 years, 7 months ago**Selected Answer: A**

You can use S3 Object Lock to store objects using a write-once-read-many (WORM) model. Object Lock can help prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. You can use S3 Object Lock to meet regulatory requirements that require WORM storage, or add an extra layer of protection against object changes and deletion. Versioning is required and automatically activated as Object Lock is enabled.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>

upvoted 31 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago**Selected Answer: A**

CORRECT

A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled.

S3 Versioning allows multiple versions of an object to be stored in the same bucket. This means that when an object is modified or deleted, the previous version is preserved. S3 Object Lock adds additional protection by allowing objects to be placed under a legal hold or retention period, during which they cannot be deleted or modified. Together, S3 Versioning and

placed under a legal hold or retention period, during which they cannot be deleted or modified. Together, S3 Versioning and S3 Object Lock can be used to meet the requirement of not allowing documents to be modified or deleted after they are stored.

upvoted 12 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG

Option B, storing the documents in an S3 bucket and configuring an S3 Lifecycle policy to archive them periodically, would not prevent the documents from being modified or deleted.

Option C, storing the documents in an S3 bucket with S3 Versioning enabled and configuring an ACL to restrict all access to read-only, would also not prevent the documents from being modified or deleted, since an ACL only controls access to the object and does not prevent it from being modified or deleted.

Option D, storing the documents on an Amazon Elastic File System (Amazon EFS) volume and accessing the data in read-only mode, would prevent the documents from being modified, but would not prevent them from being deleted.

upvoted 8 times

satyaamm Most Recent 5 months, 1 week ago

Selected Answer: A

S3 Object Lock for preventing deletion as it uses Write-Once-Read-Many (WORM) and Bucket Versioning for preserving the older files and thereby preventing modification.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: A

Ans A - 'S3 Object Lock' allows objects to be stored under legal retention regime during which they cannot be changed/deleted. Versioning helps tracking...

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the correct answer

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

"S3 Object Lock can help prevent Amazon S3 objects from being deleted or overwritten for a fixed amount of time or indefinitely. "

B is archiving which won't stop deletion

C ACL can be modified

D Sounds like there will be a write volume and a read volume which means write volume will have permissions for deletion

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

S3 Versioning ensures that all versions of an object are retained when overwritten or deleted - this prevents deletion.

S3 Object Lock can be used to apply a retention period and legal hold on objects to prevent them from being overwritten or deleted, even by users with full permissions.

Option B only archives objects on a schedule but does not prevent modification or deletion.

Option C uses ACLs which can still be overridden by users with full permissions.

Option D relies on the application to enforce mounting as read-only, which is not as robust as using S3 Object Lock.

upvoted 3 times

Subhrangsu 1 year, 8 months ago

Liked the explanation for option C.Thanks!

upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: A

S3 Versioning allows you to preserve every version of a document as it is uploaded or modified. This prevents accidental or intentional modifications or deletions of the documents.

S3 Object Lock allows you to set a retention period or legal hold on the objects, making them immutable during the specified period. This ensures that the stored documents cannot be modified or deleted, even by privileged users or administrators.

B. Configuring an S3 Lifecycle policy to archive documents periodically does not guarantee the prevention of document modification or deletion after they are stored.

C. Enabling S3 Versioning alone does not prevent modifications or deletions of objects. Configuring an ACL does not guarantee the prevention of modifications or deletions by authorized users.

D. Using EFS does not prevent modifications or deletions of the documents by users or processes with write permissions.
upvoted 3 times

Bmarodi 2 years ago

Selected Answer: A

S3 Versioning and S3 Object Lock enabled meet the requirements, hence A is correct ans.
upvoted 3 times

SilentMilli 2 years, 5 months ago

Selected Answer: A

Option A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled. This will ensure that the documents cannot be modified or deleted after they are stored, and will meet the regulatory requirement. S3 Versioning allows you to store multiple versions of an object in the same bucket, and S3 Object Lock enables you to apply a retention policy to objects in the bucket to prevent their deletion.
upvoted 3 times

career360guru 2 years, 5 months ago

Selected Answer: A

Option A. Object Lock will prevent modifications to documents
upvoted 2 times

HarryZ 2 years, 6 months ago

Why not C
upvoted 3 times

JayBee65 2 years, 5 months ago

Configure an ACL to restrict all access to read-only would be you could not write the docs to the bucket in the first place.
upvoted 3 times

Wpcorgan 2 years, 6 months ago

A is correct
upvoted 1 times

flbcobra 2 years, 7 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>
upvoted 2 times

Evangelia 2 years, 7 months ago

Selected Answer: A

aaaaaaaaa
upvoted 1 times

Evangelia 2 years, 7 months ago

aaaaaaaaa
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #86

Topic 1

A company has several web servers that need to frequently access a common Amazon RDS MySQL Multi-AZ DB instance. The company wants a secure method for the web servers to connect to the database while meeting a security requirement to rotate user credentials frequently.

Which solution meets these requirements?

- A. Store the database user credentials in AWS Secrets Manager. Grant the necessary IAM permissions to allow the web servers to access AWS Secrets Manager. **Most Voted**
- B. Store the database user credentials in AWS Systems Manager OpsCenter. Grant the necessary IAM permissions to allow the web servers to access OpsCenter.
- C. Store the database user credentials in a secure Amazon S3 bucket. Grant the necessary IAM permissions to allow the web servers to retrieve credentials and access the database.
- D. Store the database user credentials in files encrypted with AWS Key Management Service (AWS KMS) on the web server file system. The web server should be able to decrypt the files and access the database.

Correct Answer: A*Community vote distribution*

A (100%)

Comments**123jh10** **Highly Voted** 2 years, 7 months ago**Selected Answer: A**

Secrets Manager enables you to replace hardcoded credentials in your code, including passwords, with an API call to Secrets Manager to retrieve the secret programmatically. This helps ensure the secret can't be compromised by someone examining your code, because the secret no longer exists in the code. Also, you can configure Secrets Manager to automatically rotate the secret for you according to a specified schedule. This enables you to replace long-term secrets with short-term ones, significantly reducing the risk of compromise.

<https://docs.aws.amazon.com/secretsmanager/latest/userguide/intro.html>

upvoted 29 times

cookieMr **Highly Voted** 1 year, 11 months ago**Selected Answer: A**

B. SSM OpsCenter is primarily used for managing and resolving operational issues. It is not designed to securely store and manage credentials like AWS Secrets Manager

manage credentials like AWS Secrets Manager.

C. Storing credentials in an S3 bucket may provide some level of security, but it lacks the additional features and security controls offered by AWS Secrets Manager.

D. While using KMS for encryption is a good practice, managing credentials directly on the web server file system can introduce complexities and potential security risks. It can be challenging to securely manage and rotate credentials across multiple web servers, especially when considering scalability and automation.

In summary, option A is the recommended solution as it leverages AWS Secrets Manager, which is purpose-built for securely storing and managing secrets, and provides the necessary IAM permissions to allow the web servers to access the credentials securely.

upvoted 7 times

VenkatGovindu Most Recent 2 months, 3 weeks ago

Selected Answer: A

Rotate Key ==> Secrets Manager

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: A

AWS Secret's Manager is the most suitable option here as it provides rotation of credentials using Lambda.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: A

Ans A - Secrets Manager, auto rotates...

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the right answer, you should never store any of your credentials in files, even if they are encrypted. Also, secrets manager fulfills the credentials rotation condition.

upvoted 2 times

Duckydoo 1 year ago

Selected Answer: A

While I agree that A is the most feasible answer, I don't see how it satisfies the "a secure method for the web servers to connect to the database" requirement. ASM is about securely storing and rotating secrets, but has nothing to do with "secure connection" between the web servers and RDS. That would require something like IAM DB authentication, which is not even mentioned.

upvoted 3 times

CanoSys 10 months, 4 weeks ago

Yeah, It's a stupid question.

upvoted 1 times

Z0r 1 year, 1 month ago

rotation = SM

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

AWS Secrets Manager is best for storing credentials and supports auto rotation so A is the best choice

upvoted 4 times

MrPCarrot 1 year, 6 months ago

A = Rotation of user credentials can be automated using Secrets Manager.

upvoted 2 times

Buff 1 year, 7 months ago

kuttyit 1 year, 7 months ago

option A is the recommended solution as it leverages AWS Secrets Manager, which is purpose-built for securely storing and managing secrets, and provides the necessary IAM permissions to allow the web servers to access the credentials securely.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

AWS Secrets Manager to the rescue....up up and awaaaay

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

The correct answer is A.

Here is the explanation:

AWS Secrets Manager is a service that helps you store, manage, and rotate secrets. Secrets Manager is a good choice for storing database user credentials because it is secure and scalable.

IAM permissions can be used to grant web servers access to AWS Secrets Manager. This will allow the web servers to retrieve the database user credentials from Secrets Manager and use them to connect to the database.

Rotation of user credentials can be automated using Secrets Manager. This will ensure that the database user credentials are rotated on a regular basis, meeting the security requirement.

upvoted 3 times

Bmarodi 2 years ago

Selected Answer: A

Option A is ans.

upvoted 3 times

vherman 2 years, 3 months ago

Selected Answer: A

A is correct

upvoted 2 times

thensanity 2 years, 5 months ago

literally screams for AWS secrets manager to rotate the credentials

upvoted 5 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: A

CORRECT

Option A. Store the database user credentials in AWS Secrets Manager. Grant the necessary IAM permissions to allow the web servers to access AWS Secrets Manager.

Option A is correct because it meets the requirements specified in the question: a secure method for the web servers to connect to the database while meeting a security requirement to rotate user credentials frequently. AWS Secrets Manager is designed specifically to store and manage secrets like database credentials, and it provides an automated way to rotate secrets every time they are used, ensuring that the secrets are always fresh and secure. This makes it a good choice for storing and managing the database user credentials in a secure way.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG

Option B, storing the database user credentials in AWS Systems Manager OpsCenter, is not a good fit for this use case because OpsCenter is a tool for managing and monitoring systems, and it is not designed for storing and managing secrets.

Option C, storing the database user credentials in a secure Amazon S3 bucket, is not a secure option because S3 buckets are not designed to store secrets. While it is possible to store secrets in S3, it is not recommended because S3 is not a secure secrets management service and does not provide the same level of security and automation as AWS Secrets Manager.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option D, storing the database user credentials in files encrypted with AWS Key Management Service (AWS KMS) on the web server file system, is not a secure option because it relies on the security of the web server file system, which may not be as secure as a dedicated secrets management service like AWS Secrets Manager. Additionally, this option does not meet the requirement to rotate user credentials frequently because it does not provide an automated way to rotate the credentials.

upvoted 6 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #87

Topic 1

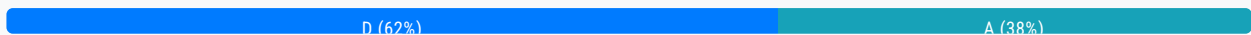
A company hosts an application on AWS Lambda functions that are invoked by an Amazon API Gateway API. The Lambda functions save customer data to an Amazon Aurora MySQL database. Whenever the company upgrades the database, the Lambda functions fail to establish database connections until the upgrade is complete. The result is that customer data is not recorded for some of the event.

A solutions architect needs to design a solution that stores customer data that is created during database upgrades. Which solution will meet these requirements?

- A. Provision an Amazon RDS proxy to sit between the Lambda functions and the database. Configure the Lambda functions to connect to the RDS proxy.
- B. Increase the run time of the Lambda functions to the maximum. Create a retry mechanism in the code that stores the customer data in the database.
- C. Persist the customer data to Lambda local storage. Configure new Lambda functions to scan the local storage to save the customer data to the database.
- D. Store the customer data in an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Create a new Lambda function that polls the queue and stores the customer data in the database. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

brushek **Highly Voted** 2 years, 8 months ago

Selected Answer: A

<https://aws.amazon.com/rds/proxy/>

RDS Proxy minimizes application disruption from outages affecting the availability of your database by automatically connecting to a new database instance while preserving application connections. When failovers occur, RDS Proxy routes requests directly to the new database instance. This reduces failover times for Aurora and RDS databases by up to 66%.

upvoted 58 times

SaurabhTiwari1 1 year, 5 months ago

The original question was about handling a situation where the database is unavailable due to an upgrade. not a failover

The original question was about renaming a database where the database is unavailable due to an upgrade, not a removal situation. During a database upgrade, the database instance is not available, and RDS Proxy would not be able to connect to a new database instance because there isn't one.

In this specific scenario, using Amazon SQS as described in option D provides a buffer for the incoming data during the period when the database is unavailable. This ensures that no data is lost, and it can be written to the database once the upgrade is complete.

upvoted 46 times

lofzee 1 year ago

so many upvotes for an incorrect answer.

the question doesn't mention having more than one DB so how does RDS proxy help here?

upvoted 11 times

PassNow1234 2 years, 5 months ago

This is MySQL Database. RDS proxy = no no

upvoted 5 times

Robrobtutu 2 years, 1 month ago

It literally says RDS Proxy is available for Aurora MySQL on the link in the comment you're replying to.

upvoted 6 times

pentium75 1 year, 5 months ago

But still RDS proxy won't help because during upgrades there is no database that it could proxy to.

upvoted 4 times

attila9778 2 years, 6 months ago

Aurora supports RDS proxy!

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 6 times

123jh10 Highly Voted 2 years, 7 months ago

Selected Answer: D

The answer is D.

RDS Proxy doesn't support Aurora DBs. See limitations at:

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 36 times

gcmrjbr 2 years, 6 months ago

You can use RDS Proxy with Aurora Serverless v2 clusters but not with Aurora Serverless v1 clusters.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 8 times

JayBee65 2 years, 5 months ago

It does, according to that link

upvoted 1 times

adeyinkaamole 1 year, 9 months ago

This not RDS supports Aurora mysql database. All the limitations listed in the link you posted above are not related to the question, hence the answer is B

upvoted 2 times

adeyinkaamole 1 year, 9 months ago

I meant the answer answer is A

upvoted 1 times

tinyfoot 2 years, 6 months ago

Actually RDS Proxy supports Aurora DBs running on PostgreSQL and MySQL.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Concepts.Aurora_Fea_Regions_DB-

eng.Feature.RDS_Proxy.html

With RDS proxy, you only expose a single endpoint for request to hit and any failure of the primary DB in a Multi-AZ configuration is will be managed automatically by RDS Proxy to point to the new primary DB. Hence RDS proxy is the most efficient way of solving the issue as additional code change is required.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.howitworks.html>

upvoted 11 times

Duke_YU 2 years, 2 months ago

The question doesn't say the RDS is deployed in a Mutli-AZ mode. which means RDS is not accessible during upgrade anyway. RDS proxy couldn't resolve the DB HA issue. The question is looking for a solution to store the data during DB upgrade. I don't know RDS proxy very well, but the RDS proxy introduction doesn't mention it has the capability of storing data. So, answer A couldn't store the data created during the DB upgrade.

I'm assuming this is a bad question design. The expected answer is A, but the question designer missed some important information.

upvoted 6 times

rismail 2 years, 1 month ago

<https://aws.amazon.com/rds/proxy/>, if you go down the page, you will see that RDS is deployed in Multi-AZ (mazon RDS Proxy is highly available and deployed over multiple Availability Zones (AZs) to protect you from infrastructure failure. Each AZ runs on its own physically distinct, independent infrastructure and is engineered to be highly reliable. In the unlikely event of an infrastructure failure, the RDS Proxy endpoint remains online and consistent allowing your application to continue to run database operations.) from the link.

upvoted 1 times

jorgemenegaz Most Recent 3 weeks, 6 days ago

Selected Answer: A

Lambda needs to be in the same network communication to aurora, you can do this using RDS Proxy.

upvoted 1 times

dattateja8 2 months, 2 weeks ago

Selected Answer: D

Ans D: This is the best solution. By using an SQS FIFO queue, you can store customer data reliably even when the database is temporarily unavailable. The Lambda function can immediately write the data to the queue when the database is not available. A second Lambda function can then poll the queue and process the customer data when the database becomes available again. This ensures that no data is lost, and processing happens in an orderly manner.

upvoted 1 times

Vandaman 3 months, 3 weeks ago

Selected Answer: D

Scenario describes 1 DB and when it's unavailable due to an upgrade. Best solution is to queue incoming data until DB is available.

upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: D

Initially I voted A. But thats wrong for one reason. RDS proxy wont help if there is no more than 1 database, but again - if there is just one database, the application is down during the downtime anyways, there is nothing that can be done. However, a SQS queue is a better option for the reason that the messages persist until marked processed. That guarantee alone is good enough to make D a better choice.

But jeez! talk about trickery!! deceptively misleading!

upvoted 2 times

zdi561 4 months, 2 weeks ago

Selected Answer: A

D does not work. because the during db upgrade the connection is disrupted for a while during which the request in SQS will be consumed and lost

upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: A

Selected Answer: A

Google does aws aurora db proxy help db upgrade, proxy will cache transaction,
upvoted 1 times

AshishDhole 4 months, 4 weeks ago

Selected Answer: D

decouple the application so that we can save the failed transactions and then pull after the upgrades complete, so D should be the correct answer (SQS Queue)

upvoted 1 times

jayessh 7 months ago

Selected Answer: D

this is not failover situation so no need to use rds proxy. i dont remember rds proxy to hold data incase for db unavailability

upvoted 1 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: A

A for sure

upvoted 2 times

lixep 8 months, 2 weeks ago

Selected Answer: D

It's not a failover situation, it's just temporarily unavailable.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans D - as excellently explained by SaurabhTiwari1 (9 mth ago) "The original question was about handling a situation where the database is unavailable due to an upgrade, not a failover situation. During a database upgrade, the database instance is not available, and RDS Proxy would not be able to connect to a new database instance because there isn't one."

In this specific scenario, using Amazon SQS as described in option D provides a buffer for the incoming data during the period when the database is unavailable. This ensures that no data is lost, and it can be written to the database once the upgrade is complete

upvoted 4 times

rpmaws 9 months, 3 weeks ago

Selected Answer: D

A is incorrect

upvoted 1 times

d7d08dc 9 months, 3 weeks ago

the answer is D:

This solution ensures that customer data is not lost during database upgrades. The data is stored in the FIFO queue until the database is available again. The new Lambda function can then process the data from the queue and store it in the database. This design provides a buffer for the customer data and decouples the data ingestion from the data processing, increasing the resilience of the system during database upgrades.

upvoted 2 times

ChinthaGurumurthi 10 months, 3 weeks ago

Selected Answer: D

By using Amazon RDS Proxy, you can allow your applications to pool and share database connections to improve their ability to scale. RDS Proxy makes applications more resilient to database failures by automatically connecting to a standby DB instance while preserving application connections. Using RDS Proxy, you can handle unpredictable surges in database traffic. Otherwise, these surges might cause issues due to oversubscribing connections or new connections being created at a fast rate.

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D is the correct answer because it offers storage for the data while the database is updating, using SQS is better than using RDS proxy, because SQS is an independent service and it decouples the system, while RDS proxy might also face some trouble connecting to the database while it's updating.

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #88

Topic 1

A survey company has gathered data for several years from areas in the United States. The company hosts the data in an Amazon S3 bucket that is 3 TB in size and growing. The company has started to share the data with a European marketing firm that has S3 buckets. The company wants to ensure that its data transfer costs remain as low as possible.

Which solution will meet these requirements?

- A. Configure the Requester Pays feature on the company's S3 bucket. **Most Voted**
- B. Configure S3 Cross-Region Replication from the company's S3 bucket to one of the marketing firm's S3 buckets.
- C. Configure cross-account access for the marketing firm so that the marketing firm has access to the company's S3 bucket.
- D. Configure the company's S3 bucket to use S3 Intelligent-Tiering. Sync the S3 bucket to one of the marketing firm's S3 buckets.

Correct Answer: A*Community vote distribution***Comments****Six_Fingered_Jose** **Highly Voted** 2 years, 7 months ago**Selected Answer: B**

this question is too vague imho

if the question is looking for a way to incur charges to the European company instead of the US company, then requester pay makes sense.

if they are looking to reduce overall data transfer cost, then B makes sense because the data does not leave the AWS network, thus data transfer cost should be lower technically?

A. makes sense because the US company saves money, but the European company is paying for the charges so there is no overall saving in cost when you look at the big picture

I will go for B because they are not explicitly stating that they want the other company to pay for the charges
upvoted 72 times

FlyingHawk 4 months, 2 weeks ago

When you use S3 Cross-Region Replication (CRR), the source bucket owner (in this case, the company sharing the data) is

when you use S3 Cross-Region Replication (CRR), the source bucket owner (in this case, the company sharing the data) is responsible for paying the data replication costs, including:

Data transfer fees: The cost of transferring data from the source S3 bucket to the destination bucket in a different AWS Region.

Replication requests: Charges for each PUT request made during replication to the destination bucket.

B minimizes the data transfer cost of the requester, the Europe company. A minimize the data transfer cost of survey company.
upvoted 2 times

FlyingHawk 4 months, 2 weeks ago

The question mentions that the consumer survey company has granted the marketing firm access to the S3 bucket. This suggests that the consumer survey company is the one managing the bucket and likely wants to minimize its own costs.

upvoted 1 times

tvvtv 6 months ago

Don't you still have to pay for cross region transfer? Yes it's lower than out-of-aws but I think there's still a cost betw regions. Maybe it's just certain services but I thought S3 was one of them.

upvoted 1 times

tonybuivannghia 8 months, 3 weeks ago

I disagree with your opinion, because S3 Cross-Region is only transfer new data to new region, the old data can't. So A is correct.

upvoted 4 times

123jh10 Highly Voted 2 years, 7 months ago

Selected Answer: A

"Typically, you configure buckets to be Requester Pays buckets when you want to share data but not incur charges associated with others accessing the data. For example, you might use Requester Pays buckets when making available large datasets, such as zip code directories, reference data, geospatial information, or web crawling data."

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/RequesterPaysBuckets.html>

upvoted 51 times

network_enthusiast Most Recent 2 months ago

Selected Answer: A

A is the correct answer

upvoted 1 times

dattateja8 2 months, 2 weeks ago

Selected Answer: A

The Requester Pays feature in Amazon S3 allows the company to charge the marketing firm for the data transfer costs when they access the company's S3 bucket. By enabling this feature, the marketing firm will be responsible for the data transfer costs when they access the data, rather than the company. This ensures that the company does not incur high data transfer costs when sharing large datasets with an external party, especially across regions.

upvoted 1 times

mc0226 3 months ago

Selected Answer: A

- In general, bucket owners pay for all Amazon S3 storage and data transfer costs associated with their bucket
- With Requester Pays buckets, the requester instead of the bucket owner pays the cost of the request and the data download from the bucket

upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: A

The question is clear and answer too. undoubtedly A - one can have "Requester pays" feature on a S3 bucket! My retention isn't great but I remember this from the S3 lesson!

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/RequesterPaysBuckets.html>

upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: A

B is not right because we do not know the usage pattern by the marketing company. copying whole data may cost much more
upvoted 1 times

FlyingHawk 4 months, 2 weeks ago

Selected Answer: A

The survey company wants to share its data transfer costs as low as possible, to replicate the data to the Europe company's account, there will still be some data transfer costs for survey company to pay, so it will be better to request the Europe's market company to pay.

upvoted 1 times

FlyingHawk 4 months, 2 weeks ago

Another reason to select A is the A can scale in the case of more than one company want their survey dataset. B cannot scale.

upvoted 1 times

FlyingHawk 4 months, 2 weeks ago

For B, When you use S3 Cross-Region Replication (CRR), the source bucket owner (in this case, the company sharing the data) is responsible for paying the data replication costs, including:

Data transfer fees: The cost of transferring data from the source S3 bucket to the destination bucket in a different AWS Region.

Replication requests: Charges for each PUT request made during replication to the destination bucket.

upvoted 1 times

chirag_a_pariikh 6 months ago

Selected Answer: B

Answer is B, cross region replication.

upvoted 1 times

Balliache520505 6 months, 2 weeks ago

Selected Answer: A

Option A. When you use S3 Cross-Region Replication (CRR), you will incur data transfer costs. These costs include a fee for transferring data between regions, which is approximately \$0.02 per GB.

upvoted 1 times

[Removed] 6 months, 2 weeks ago

Selected Answer: A

Why Not the Other Options?

B. S3 Cross-Region Replication:

Cross-Region Replication (CRR) copies data between S3 buckets in different AWS Regions but incurs data transfer costs for replication, which does not minimize costs for the survey company.

It is also unnecessary if the marketing firm can directly access the bucket.

C. Cross-Account Access:

Granting cross-account access allows the marketing firm to read data directly, but the data transfer costs would still be borne by the survey company, which contradicts the goal of minimizing the company's costs.

D. S3 Intelligent-Tiering:

S3 Intelligent-Tiering optimizes storage costs for infrequently accessed data but does not address the need to minimize data transfer costs during sharing. Syncing the bucket to the marketing firm's bucket would also incur transfer costs.

upvoted 1 times

Ode7d1b 6 months, 3 weeks ago

Selected Answer: A

The Requester Pays feature is the most cost-effective solution for this scenario, as it shifts the data transfer costs to the marketing firm while keeping the data accessible.

upvoted 1 times

Carlini2020 7 months ago

Selected Answer: A

B has actually more cost. A is the answer.

upvoted 1 times

JA2018 7 months ago

Selected Answer: A

If I understand the question correctly, the owner is already paying to maintain the data (size is still growing) in its S3 bucket. The owner wants to ensure that its data transfer costs remain as low as possible (implies that it is best if the transfer cost is \$0).

For option A, the requestor will have to bear the cost of the data transfer request from the bucket, with the data owner incurring a \$0 transfer cost, which is optimal for the data owner.

This question does not state that the original data owner must also consider the cost incurred by the requestor and then find an optimal ways to share the data transfer cost for both parties.

My 2 cents' worth.

upvoted 2 times

Neilossi 7 months ago

Selected Answer: A

The company wants to ensure that "its" data transfer costs remain as low as possible, so I choose A

upvoted 1 times

DevanshGupta 7 months, 3 weeks ago

Per my understanding, the company is already paying to maintain the data(data keeps growing) in the S3 bucket. The company wants to ensure that its data transfer costs remain as low as possible (implies that it is best if the transfer cost is \$0).

As per option A, the requestor would bear the cost of the request and the data downloaded from the bucket, causing the data owner to incur a \$0 transfer cost, which is optimal for the data owner.

The question does not say that the data owner must also consider the cost incurred by the requestor and then find an optimal cost solution.

upvoted 2 times

aturret 8 months, 1 week ago

Selected Answer: B

No idea but I guess B. request payer sounds like stpd

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #89

Topic 1

A company uses Amazon S3 to store its confidential audit documents. The S3 bucket uses bucket policies to restrict access to audit team IAM user credentials according to the principle of least privilege. Company managers are worried about accidental deletion of documents in the S3 bucket and want a more secure solution.

What should a solutions architect do to secure the audit documents?

- A. Enable the versioning and MFA Delete features on the S3 bucket. **Most Voted**
- B. Enable multi-factor authentication (MFA) on the IAM user credentials for each audit team IAM user account.
- C. Add an S3 Lifecycle policy to the audit team's IAM user accounts to deny the s3:DeleteObject action during audit dates.
- D. Use AWS Key Management Service (AWS KMS) to encrypt the S3 bucket and restrict audit team IAM user accounts from accessing the KMS key.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jhl0 **Highly Voted** 2 years, 7 months ago

Selected Answer: A

Same as Question #44

upvoted 16 times

satyaamm **Most Recent** 5 months, 1 week ago

Selected Answer: A

Using MFA Delete is the most suitable option here as it avoids deletion of files.

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the correct answer.

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

Accidental deletion is the key. Deletion is allowed but MFA deletion ensures that deletion requires an additional step.
<https://docs.aws.amazon.com/AmazonS3/latest/userguide/MultiFactorAuthenticationDelete.html>

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

Enable the versioning to ensure restoration in case of accidental deletion and MFA Delete for double verification before deletion.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

Versioning will keep multiple variants of an object in case one is accidentally or intentionally deleted - the previous versions can still be restored.

MFA Delete requires additional authentication to permanently delete an object version. This prevents accidental deletion

upvoted 3 times

cookieMr 1 year, 11 months ago

B. Enabling MFA on the IAM user credentials adds an extra layer of security to the user authentication process. However, it does not specifically address the concern of accidental deletion of documents in the S3 bucket.

C. Adding an S3 Lifecycle policy to deny the delete action during audit dates would prevent intentional deletions during specific time periods. However, it does not address accidental deletions that can occur at any time.

D. Using KMS for encryption and restricting access to the KMS key provides additional security for the data stored in the S3 . However, it does not directly prevent accidental deletion of documents in the S3.

Enabling versioning and MFA Delete on the S3 (option A) is the most appropriate solution for securing the audit documents. Versioning ensures that multiple versions of the documents are stored, allowing for easy recovery in case of accidental deletions. Enabling MFA Delete requires the use of multi-factor authentication to authorize deletion actions, adding an extra layer of protection against unintended deletions.

upvoted 3 times

beginnercloud 2 years ago

Selected Answer: A

A is answer.

upvoted 1 times

Bmarodi 2 years ago

Selected Answer: A

A is answer.

upvoted 1 times

Robrobtutu 2 years, 1 month ago

Selected Answer: A

A is correct.

upvoted 1 times

remand 2 years, 4 months ago

Selected Answer: A

only accidental deletion should be avoided. IAM policy will completely remove their access.hence, MFA is the right choice.

upvoted 2 times

karbob 2 years, 5 months ago

what about : IAM policies are used to specify permissions for AWS resources, and they can be used to allow or deny specific actions on those resources.

```
{  
"Version": "2012-10-17",
```

```
"Statement": [
{
  "Sid": "DenyDeleteObject",
  "Effect": "Deny",
  "Action": "s3:DeleteObject",
  "Resource": [
    "arn:aws:s3::my-bucket/my-object",
    "arn:aws:s3::my-bucket"
  ]
}
]
```

upvoted 2 times

remand 2 years, 4 months ago

only accidental deletion should be avoided. IAM policy will completely remove their access.hence, MFA is the right choice.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: A

The solution architect should do Option A: Enable the versioning and MFA Delete features on the S3 bucket.

This will secure the audit documents by providing an additional layer of protection against accidental deletion. With versioning enabled, any deleted or overwritten objects in the S3 bucket will be preserved as previous versions, allowing the company to recover them if needed. With MFA Delete enabled, any delete request made to the S3 bucket will require the use of an MFA code, which provides an additional layer of security.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option B: Enable multi-factor authentication (MFA) on the IAM user credentials for each audit team IAM user account, would not provide protection against accidental deletion.

Option C: Adding an S3 Lifecycle policy to the audit team's IAM user accounts to deny the s3:DeleteObject action during audit dates, which would not provide protection against accidental deletion outside of the specified audit dates.

Option D: Use AWS Key Management Service (AWS KMS) to encrypt the S3 bucket and restrict audit team IAM user accounts from accessing the KMS key, would not provide protection against accidental deletion.

upvoted 3 times

career360guru 2 years, 5 months ago

Selected Answer: A

A is the right answer

upvoted 1 times

Wpcorgan 2 years, 6 months ago

A is correct

upvoted 1 times

Jtic 2 years, 7 months ago

Selected Answer: A

Enable the versioning and MFA Delete features on the S3 bucket.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #90

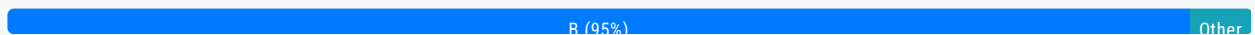
Topic 1

A company is using a SQL database to store movie data that is publicly accessible. The database runs on an Amazon RDS Single-AZ DB instance. A script runs queries at random intervals each day to record the number of new movies that have been added to the database. The script must report a final total during business hours. The company's development team notices that the database performance is inadequate for development tasks when the script is running. A solutions architect must recommend a solution to resolve this issue. Which solution will meet this requirement with the LEAST operational overhead?

- A. Modify the DB instance to be a Multi-AZ deployment.
- B. Create a read replica of the database. Configure the script to query only the read replica. **Most Voted**
- C. Instruct the development team to manually export the entries in the database at the end of each day.
- D. Use Amazon ElastiCache to cache the common queries that the script runs against the database.

Correct Answer: B

Community vote distribution



Comments

alvarez100 **Highly Voted** 2 years, 8 months ago

Selected Answer: B

Elasti Cache if for reading common results. The script is looking for new movies added. Read replica would be the best choice.
upvoted 45 times

Gil80 **Highly Voted** 2 years, 7 months ago

Selected Answer: B

- You have a production DB that is taking on a normal load
- You want to run a reporting application to run some analytics
- You create a read replica to run the new workload there
- The prod application is unaffected
- Read replicas are used for SELECT (=read) only kind of statements

Therefore I believe B to be the better answer.

As for "D" - ElastiCache use cases are:

1. Your data is slow or expensive to get when compared to cache retrieval.
2. Users access your data often

2. Users access your data often.

3. Your data stays relatively the same, or if it changes quickly staleness is not a large issue.

1 - Somewhat true.

2 - Not true for our case.

3 - Also not true. The data changes throughout the day.

For my understanding, caching has to do with millisecond results, high-performance reads. These are not the issues mentioned in the questions, therefore B.

upvoted 18 times

NitiATOS 2 years, 4 months ago

I will support this by point to the question : " with the LEAST operational overhead?"

Configuring the read replica is much easier than configuring and integrating new service.

upvoted 5 times

satyaamm Most Recent 5 months, 1 week ago

Selected Answer: B

Read replicas are replications of original databases and are the most suitable option for running scripts here.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: B

Ans B - as per Gil80 (1 year, 10mth ago)

"Selected Answer: B

- You have a production DB that is taking on a normal load
- You want to run a reporting application to run some analytics
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As for "D" - ElastiCache use cases are:

1. Your data is slow or expensive to get when compared to cache retrieval.
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3. Your data stays relatively the same, or if it changes quickly staleness is not a large issue.

1 - Somewhat true.

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3 - Also not true. The data changes throughout the day.

For my understanding, caching has to do with millisecond results, high-performance reads. These are not the issues mentioned in the questions, therefore B."

upvoted 3 times

jaradat02 10 months, 3 weeks ago

Selected Answer: B

option B for sure, read replica is designed for this very use case, improving performance, on the other hand, enabling multi-AZ improves availability and not performance.

upvoted 2 times

MehulKapadia 1 year, 2 months ago

Selected Answer: B

A. Having multi-AZ database would increase availability, bu not performance.

C. Not practical. Huge Operational Overhead. (Solution should be LEAST operational overhead)

D. Good for fixed queries with fixed results. not a good fit in this case as script is looking for new results in DB. It has to scan the database.

Correct Answer B. Read replica ensure you have dedicated read instance with its own resources.

upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

Just read from read replica.

A: This will make it HA but won't solve any problems

C: We want an AWS solution not change the development team's ways of working

D: Elasticache is cache of read queries when data doesn't change. It's useless for finding new data.

upvoted 3 times

bujuman 1 year, 5 months ago

Selected Answer: A

Answer C is inconceivable according to LEAST operational overhead?

We will exclude answer D because question is about RDS databases and ElastiCache is not.

Between answers A and B, A is the most appropriate answer due to 2 following points:

- Possible to transform a Single-AZ RDS to Multi-AZ
- LEAST operational overhead

upvoted 2 times

smdrouiss 1 year, 6 months ago

Selected Answer: A

It is A, because the scenario mentions "single db instance" which is not possible to enable read replica

upvoted 1 times

pentium75 1 year, 5 months ago

Doesn't it become a multi-instance DB when you add a read replica? A can't be because you can't read from the passive replica of a multi-AZ DB.

upvoted 3 times

slimen 1 year, 7 months ago

Selected Answer: B

lol seriously the person who wrote the answer wants us to fail

upvoted 7 times

tom_cruise 1 year, 8 months ago

Selected Answer: B

This is what we do in the real world.

upvoted 1 times

joshik 1 year, 8 months ago

Selected Answer: B

- Cached data might not always be up-to-date, so you need to manage cache expiry and invalidation carefully.
- It may require some code changes to implement caching logic in your script.
- ElastiCache comes with additional costs, so you should assess the cost implications based on your usage.

upvoted 2 times

underdogpex 1 year, 9 months ago

Selected Answer: B

Why not D:

While ElastiCache can be relatively easy to set up, it still requires ongoing management, monitoring, and potentially scaling as the dataset and query load grow. This introduces operational overhead that may not align with the goal of minimizing operational work.

upvoted 2 times

Router 1 year, 9 months ago

the correct answer should be A, you can't create a read replica on a single-AZ DB instance

upvoted 1 times

effiecancode 11 months, 1 week ago

Actually you can

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

a read replica is always fit for these type of scenarios.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

The key requirements are:

The script must report a final total during business hours

Resolve the issue of inadequate database performance for development tasks when the script is running

With the least operational overhead

upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: B

A. Modifying the DB to be a Multi-AZ deployment improves high availability and fault tolerance but does not directly address the performance issue during the script execution.

C. Instructing the development team to manually export the entries in the database introduces manual effort and is not a scalable or efficient solution.

D. While using ElastiCache for caching can improve read performance for common queries, it may not be the most suitable solution for the scenario described. Caching is effective for reducing the load on the database for frequently accessed data, but it may not directly address the performance issue during the script execution.

Creating a read replica of the database (option B) provides a scalable solution that offloads read traffic from the primary database. The script can be configured to query the read replica, reducing the impact on the primary database during the script execution.

upvoted 5 times

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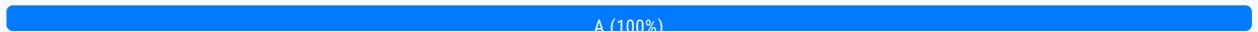
Question #91

Topic 1

A company has applications that run on Amazon EC2 instances in a VPC. One of the applications needs to call the Amazon S3 API to store and read objects. According to the company's security regulations, no traffic from the applications is allowed to travel across the internet.

Which solution will meet these requirements?

- A. Configure an S3 gateway endpoint. **Most Voted**
- B. Create an S3 bucket in a private subnet.
- C. Create an S3 bucket in the same AWS Region as the EC2 instances.
- D. Configure a NAT gateway in the same subnet as the EC2 instances.

Correct Answer: A*Community vote distribution*A (100%)**Comments****ArielSchivo** **Highly Voted** 2 years, 7 months ago**Selected Answer: A**

Gateway endpoints provide reliable connectivity to Amazon S3 and DynamoDB without requiring an internet gateway or a NAT device for your VPC. It should be option A.

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 33 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago**Selected Answer: A**

CORRECT

The correct solution is Option A (Configure an S3 gateway endpoint.)

A gateway endpoint is a VPC endpoint that you can use to connect to Amazon S3 from within your VPC. Traffic between your VPC and Amazon S3 never leaves the Amazon network, so it doesn't traverse the internet. This means you can access Amazon S3 without the need to use a NAT gateway or a VPN connection.

WRONG

Option B (creating an S3 bucket in a private subnet) is not a valid solution because S3 buckets do not have subnets.

Option C (creating an S3 bucket in the same AWS Region as the EC2 instances) is not a requirement for meeting the given security regulations.

Option D (configuring a NAT gateway in the same subnet as the EC2 instances) is not a valid solution because it would allow traffic to leave the VPC and travel across the Internet.

upvoted 19 times

PaulGa Most Recent 9 months ago

Selected Answer: A

Ans A - S3 gateway endpoint: dedicated end-end and private

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the correct answer

upvoted 1 times

effiecancode 11 months, 1 week ago

it's definitely A

upvoted 1 times

JohnZh 1 year, 2 months ago

A. Configure an S3 gateway endpoint.

Correct: <https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC.

Additional you need to configure the route table for the subnet that EC2 stays, but we have the key word here.

B. Create an S3 bucket in a private subnet.

I am not aware that we can create S3 bucket in certain subnet.

C. Create an S3 bucket in the same AWS Region as the EC2 instances.

Not enough. Without VPC gateway endpoint, access will through go out to the internet.

D. Configure a NAT gateway in the same subnet as the EC2 instances.

NAT gateway outbound traffic should also go out to the internet.

upvoted 2 times

Charumathi 1 year, 4 months ago

Selected Answer: A

You can access Amazon S3 from your VPC using gateway VPC endpoints. After you create the gateway endpoint, you can add it as a target in your route table for traffic destined from your VPC to Amazon S3.

There is no additional charge for using gateway endpoints.

Amazon S3 supports both gateway endpoints and interface endpoints. With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC, and with no additional cost. However, gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway. For those scenarios, you must use an interface endpoint, which is available for an additional cost.

<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

EC2 to S3 without public interne = S3 gatwat

B: Cannot be implemented

C: Even if you create EC2 and S3 in same region, without a S3 gateway it will use the public internet

D: Makes no sense, NAT gateway in the subnet as EC2 instance to do what?

upvoted 2 times

Ruffyt 1 year, 7 months ago

A gateway endpoint is a VPC endpoint that you can use to connect to Amazon S3 from within your VPC. Traffic between your

A gateway endpoint is a VPC endpoint that you can use to connect to Amazon S3 from within your VPC. Traffic between your VPC and Amazon S3 never leaves the Amazon network, so it doesn't traverse the internet. This means you can access Amazon S3 without the need to use a NAT gateway or a VPN connection

upvoted 2 times

David_Ang 1 year, 8 months ago

Selected Answer: A

Answer "A" is correct because an endpoint create a way for the data to travel in the VPC

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

Prevent traffic from traversing the internet = Gateway VPC endpoint for S3.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

Configure an S3 gateway endpoint

upvoted 1 times

tamefi5512 1 year, 11 months ago

Selected Answer: A

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 2 times

cookieMr 1 year, 11 months ago

B. Creating an S3 in a private subnet restricts direct internet access to the bucket but does not provide a direct and secure connection between the EC2 and the S3. The application would still need to traverse the internet to access the S3 API.

C. Creating an S3 in the same Region as the EC2 does not inherently prevent traffic from traversing the internet.

D. Configuring a NAT gateway allows outbound internet connectivity for resources in private subnets, but it does not provide a direct and secure connection to the S3 service. The traffic from the EC2 to the S3 API would still traverse the internet.

The most suitable solution is to configure an S3 gateway endpoint (option A). It provides a secure and private connection between the VPC and the S3 service without requiring the traffic to traverse the internet. With an S3 gateway endpoint, the EC2 can access the S3 API directly within the VPC, meeting the security requirement of preventing traffic from traveling across the internet.

upvoted 3 times

Bmarodi 2 years ago

Selected Answer: A

Configure an S3 gateway endpoint is answer.

upvoted 2 times

gustavtd 2 years, 5 months ago

Selected Answer: A

S3 Gateway Endpoint is a VPC endpoint,

upvoted 2 times

langiac 2 years, 6 months ago

Selected Answer: A

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #92

Topic 1

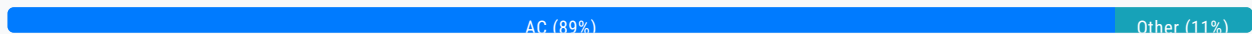
A company is storing sensitive user information in an Amazon S3 bucket. The company wants to provide secure access to this bucket from the application tier running on Amazon EC2 instances inside a VPC.

Which combination of steps should a solutions architect take to accomplish this? (Choose two.)

- A. Configure a VPC gateway endpoint for Amazon S3 within the VPC. **Most Voted**
- B. Create a bucket policy to make the objects in the S3 bucket public.
- C. Create a bucket policy that limits access to only the application tier running in the VPC. **Most Voted**
- D. Create an IAM user with an S3 access policy and copy the IAM credentials to the EC2 instance.
- E. Create a NAT instance and have the EC2 instances use the NAT instance to access the S3 bucket.

Correct Answer: AC

Community vote distribution



Comments

ArielSchivo **Highly Voted** 2 years, 7 months ago

Selected Answer: AC

Options A and C.

<https://aws.amazon.com/premiumsupport/knowledge-center/s3-private-connection-no-authentication/>
upvoted 5 times

awsgeek75 **Highly Voted** 1 year, 5 months ago

Selected Answer: AC

A: VPC S3 gateway for direct connection (no public internet) to access S3
C: Bucket policy to secure access and only allow the VPC application tier to access it

B: Opens up to public

D: Not secure to copy credentials

E: NAT instance (obsolete now) is not useful for limiting resource access, it's for subnet connections

upvoted 5 times

satyaamm **Most Recent** 5 months, 1 week ago

Selected Answer: AC

VPC Gateway endpoint provides secure access to S3 and DynamoDB while the bucket policy allows access only to the application tier making it a secure connection.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: AD

A,D - altho' I stand corrected: D means copying credentials which introduces a security risk... so that means A,C

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: AC

A removes the need for a NAT gateway and keeps the connection private, C restricts access to the bucket.

upvoted 3 times

rityoui 1 year, 5 months ago

no one mentioned the translation issue, "limit access to sth" sounds like limit this but allow others, confusing for non-English speaker.

upvoted 3 times

Ruffyit 1 year, 7 months ago

) Configure a VPC gateway endpoint for Amazon S3 within the VPC.

C) Create a bucket policy that limits access to only the application tier running in the VPC.

The key requirements are secure access to the S3 bucket from EC2 instances in the VPC.

A VPC endpoint for S3 allows connectivity from the VPC to S3 without needing internet access. The bucket policy should limit access only to the VPC by whitelisting the VPC endpoint.

upvoted 2 times

David_Ang 1 year, 8 months ago

Selected Answer: AC

These are correct because "A" and "C" ensure secure access and secure connectivity between the S3 and the EC2 instances

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AC

The key requirements are to provide secure access to the S3 bucket only from the application tier EC2 instances inside the VPC.

A VPC gateway endpoint allows private access to S3 from within the VPC without needing internet access. This keeps the traffic secure within the AWS network.

The bucket policy should limit access to only the application tier, not make the objects public. This restricts access to the sensitive data to only the authorized application tier.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AC

The correct options are:

A) Configure a VPC gateway endpoint for Amazon S3 within the VPC.

C) Create a bucket policy that limits access to only the application tier running in the VPC.

The key requirements are secure access to the S3 bucket from EC2 instances in the VPC.

A VPC endpoint for S3 allows connectivity from the VPC to S3 without needing internet access. The bucket policy should limit access only to the VPC by whitelisting the VPC endpoint.

upvoted 3 times

sohailn 1 year, 10 months ago

ac is the correct answer, as per my knowledge people are confused with IAM user we can use IAM role for secure access.
upvoted 2 times

tamefi5512 1 year, 11 months ago

Selected Answer: AC

AC is the right answer
upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: AC

- A. This eliminates the need for the traffic to go over the internet, providing an added layer of security.
- B. It is important to restrict access to the bucket and its objects only to authorized entities.
- C. This helps maintain the confidentiality of the sensitive user information by limiting access to authorized resources.
- D. In this case, since the EC2 instances are accessing the S3 bucket from within the VPC, using IAM user credentials is unnecessary and can introduce additional security risks.
- E. a NAT instance to access the S3 bucket adds unnecessary complexity and overhead.

In summary, the recommended steps to provide secure access to the S3 from the application tier running on EC2 inside a VPC are to configure a VPC gateway endpoint for S3 within the VPC (option A) and create a bucket policy that limits access to only the application tier running in the VPC (option C).
upvoted 3 times

Bmarodi 2 years ago

Selected Answer: AC

A & C the correct solutions.
upvoted 2 times

TillieEhaung 2 years, 1 month ago

Selected Answer: AC

A and C
upvoted 1 times

annabellehiro 2 years, 2 months ago

Selected Answer: AC

A and C
upvoted 1 times

Help2023 2 years, 3 months ago

Selected Answer: AC

The key part that many miss out on is 'Combination'
The other answers are not wrong but
A works with C and not with the rest as they need an internet connection.
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #93

Topic 1

A company runs an on-premises application that is powered by a MySQL database. The company is migrating the application to AWS to increase the application's elasticity and availability.

The current architecture shows heavy read activity on the database during times of normal operation. Every 4 hours, the company's development team pulls a full export of the production database to populate a database in the staging environment. During this period, users experience unacceptable application latency. The development team is unable to use the staging environment until the procedure completes.

A solutions architect must recommend replacement architecture that alleviates the application latency issue. The replacement architecture also must give the development team the ability to continue using the staging environment without delay.

Which solution meets these requirements?

A. Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Populate the staging database by implementing a backup and restore process that uses the mysqldump utility.

B. Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand. **Most Voted**

C. Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Use the standby instance for the staging database.

D. Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Populate the staging database by implementing a backup and restore process that uses the mysqldump utility.

Correct Answer: B

Community vote distribution

B (91%)

C (9%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 1 year, 11 months ago

Selected Answer: B

The recommended solution is Option B: Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand.

To alleviate the application latency issue, the recommended solution is to use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production, and use database cloning to create the staging database on-demand. This allows the development team to continue using the staging environment without delay, while also providing elasticity and availability for the

production application.

Therefore, Options A, C, and D are not recommended

upvoted 25 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Option A: Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Populating the staging database by implementing a backup and restore process that uses the mysqldump utility is not the recommended solution because it involves taking a full export of the production database, which can cause unacceptable application latency.

Option C: Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Using the standby instance for the staging database is not the recommended solution because it does not give the development team the ability to continue using the staging environment without delay. The standby instance is used for failover in case of a production instance failure, and it is not intended for use as a staging environment.

upvoted 19 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Option D: Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Populating the staging database by implementing a backup and restore process that uses the mysqldump utility is not the recommended solution because it involves taking a full export of the production database, which can cause unacceptable application latency.

upvoted 9 times

MutiverseAgent 1 year, 5 months ago

Agree, solution it seems to be the B)

1) Because the company wants "elasticity and availability" as the question mentioned, so I think this leaves us in the two questions related to Aurora discarding the RDS MySQL solution.

2) According AWS documentation

(<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.Clone.html>)

"Aurora cloning is especially useful for quickly setting up test environments using your production data, without risking data corruption"

upvoted 8 times

arashjs1993 **Highly Voted** 1 year, 1 month ago

Selected Answer: B

Aurora MySQL is very fast in comparison to RDS for creating a clone of DB, you can create a even clone of a clone while you still work on your own clone, this will allow the dev team continue working during cloning step.

<https://aws.amazon.com/blogs/aws/amazon-aurora-fast-database-cloning/>

upvoted 11 times

Rcosmos **Most Recent** 5 months ago

Selected Answer: B

Conclusão:

A configuração B com o Amazon Aurora MySQL oferece uma solução eficiente, escalável e de alto desempenho, eliminando o problema de latência do aplicativo e permitindo que a equipe de desenvolvimento use o ambiente de preparo sem atrasos. A clonagem de banco de dados Aurora é a chave para atender a esses requisitos com o mínimo impacto operacional.

upvoted 1 times

MehulKapadia 8 months, 1 week ago

Selected Answer: B

Answer B:

Few Points about Aurora Database Cloning:

- Create a new Aurora DB Cluster from an existing one
- Faster than snapshot & restore
- Uses copy-on-write protocol
- Very fast & cost-effective
- Useful to create a "staging" database from a "production" database without impacting the production database

upvoted 2 times

ronntsai127 8 months, 2 weeks ago

When the question did not say about "cost efficient", always choose Aurora MySQL > RDS MySQL, because AWS can earn more money in Aurora

upvoted 3 times

awsgeek75 11 months ago

Selected Answer: B

I'll go for B

AD: looks time consuming as mysqldump is like a table dump

C: You cannot use a standby for anything apart from read-only database. This would be an option if dev team was specifically using it for read-only mode.

<https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/>

upvoted 2 times

Ruffyt 1 year, 1 month ago

B. With Aurora, you can create a clone of the production database quickly and efficiently, without the need for time-consuming backup and restore processes. The development team can spin up the staging database on-demand, eliminating delays and allowing them to continue using the staging environment without interruption.

upvoted 2 times

Modulopi 1 year, 2 months ago

Selected Answer: B

B is the correct

upvoted 1 times

TariqKipkemei 1 year, 3 months ago

Selected Answer: C

No mention of cost, so technically both options B & C would work.

C. <https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/#:~:text=read%20replicas,-Amazon%20RDS,-now%20offers%20Multi>

B. <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.Clone.html#:~:text=cloning%20works,-Aurora%20cloning,-is%20especially%20useful>

upvoted 2 times

Guru4Cloud 1 year, 4 months ago

Selected Answer: B

Option B is the best solution that meets all the requirements:

Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand.

The key requirements are to:

Alleviate application latency caused by database exports

Give development immediate access to a staging environment

Aurora Multi-AZ replicas improves availability and provides fast failover.

Database cloning creates an instantly available copy of the production database that can be used for staging. This avoids any export or restoration del

upvoted 2 times

cookieMr 1 year, 5 months ago

Selected Answer: B

A. Populating the staging database through a backup and restore process using the mysqldump utility would still result in delays and impact application latency.

B. With Aurora, you can create a clone of the production database quickly and efficiently, without the need for time-consuming backup and restore processes. The development team can spin up the staging database on-demand, eliminating delays and allowing them to continue using the staging environment without interruption.

C. Using the standby instance for the staging database would not provide the development team with the ability to use the staging environment without delay. The standby instance is designed for failover purposes and may not be readily available for immediate use.

D. Relying on a backup and restore process using the mysqldump utility would still introduce delays and impact application latency during the data population phase.

latency during the data population phase.

upvoted 3 times

linux_admin 1 year, 8 months ago

Selected Answer: B

With Amazon Aurora MySQL, creating a staging database using database cloning is an easy process. Using database cloning will eliminate the performance issues that occur when a full export is done, and the new database is created. In addition, Amazon Aurora's high availability is provided through Multi-AZ deployment, and read replicas can be used to serve the heavy read traffic without affecting the production database. This solution provides better scalability, elasticity, and availability than the current architecture.

upvoted 5 times

alexiscloud 1 year, 8 months ago

Answer B:

upvoted 1 times

bdp123 1 year, 10 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/aws/amazon-aurora-fast-database-cloning/>

upvoted 3 times

john2323 1 year, 10 months ago

Selected Answer: B

Database cloning is the best answer

upvoted 1 times

techhb 1 year, 11 months ago

Selected Answer: B

Database cloning is right answer here.

upvoted 1 times

career360guru 1 year, 12 months ago

Option B is right.

You can not access Standby instance for Read in RDS Multi-AZ Deployments.

upvoted 4 times

aadi7 1 year, 11 months ago

This is correct, stand by instances cannot be used for read/write and is for failover targets. Read Replicas can be used for that so B is correct.

upvoted 3 times

aadi7 1 year, 11 months ago

In a RDS Multi-AZ deployment, you can use the standby instance for read-only purposes, such as running queries and reporting. This is known as a "read replica." You can create one or more read replicas of a DB instance and use them to offload read traffic from the primary instance.

<https://aws.amazon.com/about-aws/whats-new/2018/01/amazon-rds-read-replicas-now-support-multi-az-deployments/>

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #94

Topic 1

A company is designing an application where users upload small files into Amazon S3. After a user uploads a file, the file requires one-time simple processing to transform the data and save the data in JSON format for later analysis. Each file must be processed as quickly as possible after it is uploaded. Demand will vary. On some days, users will upload a high number of files. On other days, users will upload a few files or no files. Which solution meets these requirements with the LEAST operational overhead?

- A. Configure Amazon EMR to read text files from Amazon S3. Run processing scripts to transform the data. Store the resulting JSON file in an Amazon Aurora DB cluster.
- B. Configure Amazon S3 to send an event notification to an Amazon Simple Queue Service (Amazon SQS) queue. Use Amazon EC2 instances to read from the queue and process the data. Store the resulting JSON file in Amazon DynamoDB.
- C. Configure Amazon S3 to send an event notification to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function to read from the queue and process the data. Store the resulting JSON file in Amazon DynamoDB.
- D. Configure Amazon EventBridge (Amazon CloudWatch Events) to send an event to Amazon Kinesis Data Streams when a new file is uploaded. Use an AWS Lambda function to consume the event from the stream and process the data. Store the resulting JSON file in an Amazon Aurora DB cluster.

Most Voted

Correct Answer: C

Community vote distribution

C (100%)

Comments

rjam Highly Voted 2 years, 7 months ago

Option C
Dynamo DB is a NoSQL-JSON supported
upvoted 17 times

rjam 2 years, 7 months ago

also Use an AWS Lambda - serverless - less operational overhead
upvoted 12 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: C

- A. Configuring EMR and an Aurora DB cluster for this use case would introduce unnecessary complexity and operational overhead. EMR is typically used for processing large datasets and running big data frameworks like Apache Spark or Hadoop.
- B. While using S3 event notifications and SQS for decoupling is a good approach, using EC2 to process the data would introduce operational overhead in terms of managing and scaling the EC2.
- D. Using EventBridge and Kinesis Data Streams for this use case would introduce additional complexity and operational overhead compared to the other options. EventBridge and Kinesis are typically used for real-time streaming and processing of large volumes of data.

In summary, option C is the recommended solution as it provides a serverless and scalable approach for processing uploaded files using S3 event notifications, SQS, and Lambda. It offers low operational overhead, automatic scaling, and efficient handling of varying demand. Storing the resulting JSON file in DynamoDB aligns with the requirement of saving the data for later analysis.

upvoted 10 times

Dharmarajan **Most Recent** 4 months, 2 weeks ago

Selected Answer: C

C is the simplest and most widely used architecture, for example, processing photos to generate thumbnails.
upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: C

C is most suitable here as we need DynamoDB for low latency, AWS Lambda for scaling as per the demand.
upvoted 1 times

PaulGa 9 months ago

Selected Answer: C

Ans C - as per cookieMr (1 yr, 2 mth ago)

"...In summary, option C is the recommended solution as it provides a serverless and scalable approach for processing uploaded files using S3 event notifications, SQS, and Lambda. It offers low operational overhead, automatic scaling, and efficient handling of varying demand. Storing the resulting JSON file in DynamoDB aligns with the requirement of saving the data for later analysis."

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: C

Option C, fulfills the least operational overhead condition.
upvoted 2 times

TilTil 1 year, 2 months ago

Selected Answer: C

B where we use EC2 instances for processing would be ideal in situations where runtime is > 15 minutes. However the question mentions 'simple processing', hence we go for Lambda.
upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

LEAST operational overhead

A: EMR is massive programming effort for this

B: EC2 is considerable overhead

D: Nice solution but why would you use Kinesis as there is no streaming scenario here

C: Simplest and all managed services so least operational overhead compared to other options

upvoted 2 times

Ruffyit 1 year, 7 months ago

Option C is the best solution that meets the requirements with the least operational overhead:

Configure Amazon S3 to send event notification to SQS queue
Use Lambda function triggered by SQS to process each file

Use Lambda function triggered by SQS to process each file

Store output JSON in DynamoDB

This leverages serverless components like S3, SQS, Lambda, and DynamoDB to provide automated file processing without needing to provision and manage servers.

SQS queues the notifications and Lambda scales automatically to handle spikes and drops in file uploads. No EMR cluster or EC2 Fleet is needed to manage.

upvoted 2 times

Modulopi 1 year, 8 months ago

Selected Answer: C

C: Lambdas are made for that

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

C is best

upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

Option C is the best solution that meets the requirements with the least operational overhead:

Configure Amazon S3 to send event notification to SQS queue

Use Lambda function triggered by SQS to process each file

Store output JSON in DynamoDB

This leverages serverless components like S3, SQS, Lambda, and DynamoDB to provide automated file processing without needing to provision and manage servers.

SQS queues the notifications and Lambda scales automatically to handle spikes and drops in file uploads. No EMR cluster or EC2 Fleet is needed to manage.

upvoted 2 times

beginnercloud 2 years ago

Selected Answer: C

Option C is correct - Dynamo DB is a NoSQL-JSON supported

upvoted 1 times

Abrar2022 2 years ago

Selected Answer: C

SQS + LAMDA + JSON >>>>> Dynamo DB

upvoted 2 times

Bmarodi 2 years ago

Selected Answer: C

The option C is right answer.

upvoted 1 times

jy190 2 years, 1 month ago

can someone explain why SQS? it's a poll-based messaging, does it guarantee reacting the event asap?

upvoted 1 times

Zerotn3 2 years, 5 months ago

Selected Answer: C

Dynamo DB is a NoSQL-JSON supported

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

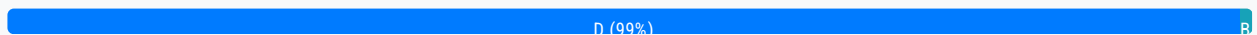
Question #95

Topic 1

An application allows users at a company's headquarters to access product data. The product data is stored in an Amazon RDS MySQL DB instance. The operations team has isolated an application performance slowdown and wants to separate read traffic from write traffic. A solutions architect needs to optimize the application's performance quickly.

What should the solutions architect recommend?

- A. Change the existing database to a Multi-AZ deployment. Serve the read requests from the primary Availability Zone.
- B. Change the existing database to a Multi-AZ deployment. Serve the read requests from the secondary Availability Zone.
- C. Create read replicas for the database. Configure the read replicas with half of the compute and storage resources as the source database.
- D. Create read replicas for the database. Configure the read replicas with the same compute and storage resources as the source database. **Most Voted**

Correct Answer: D*Community vote distribution***Comments****Buruguduystunstugudunstuy** **Highly Voted** 2 years, 5 months ago**Selected Answer: D**

The solutions architect should recommend option D: Create read replicas for the database. Configure the read replicas with the same compute and storage resources as the source database.

Creating read replicas allows the application to offload read traffic from the source database, improving its performance. The read replicas should be configured with the same compute and storage resources as the source database to ensure that they can handle the read workload effectively.

upvoted 29 times

cookieMr **Highly Voted** 1 year, 11 months ago**Selected Answer: D**

A. In a Multi-AZ deployment, a standby replica of the database is created in a different AZ for high availability and automatic failover purposes. However, serving read requests from the primary AZ alone would not effectively separate read and write traffic. Both read and write traffic would still be directed to the primary database instance, which might not fully optimize performance.

B. The secondary instance in a Multi-AZ deployment is intended for failover and backup purposes, not for actively serving read traffic. It operates in a standby mode and is not optimized for handling read queries efficiently.

C. Configuring the read replicas with half of the compute and storage resources as the source database might not be optimal. It's generally recommended to configure the read replicas with the same compute and storage resources as the source database to ensure they can handle the read workload effectively.

D. Configuring the read replicas with the same compute and storage resources as the source database ensures that they can handle the read workload efficiently and provide the required performance boost.

upvoted 8 times

Dharmarajan Most Recent 4 months, 2 weeks ago

Selected Answer: D

The read replica should be created with same capacity so it serves as a failover site as well, improving the availability of the database.

upvoted 1 times

Rcosmos 5 months ago

Selected Answer: D

A criação de réplicas de leitura com os mesmos recursos de computação e armazenamento que o banco de dados de origem resolve o problema de forma eficiente, pois:

Tráfego de leitura separado:

Direcionar o tráfego de leitura para as réplicas reduz a carga na instância primária, otimizando o desempenho geral. Manutenção da performance:

Garantir que as réplicas tenham os mesmos recursos evita gargalos, mesmo sob tráfego intenso. Implementação rápida:

A configuração de réplicas de leitura no Amazon RDS é rápida e gerenciada, permitindo uma solução quase imediata. Essa abordagem atende à necessidade da empresa de otimizar o desempenho do aplicativo de forma rápida e escalável.

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: D

Multi-AZ's offer availability but Read Replicas offers read performance.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans D - can't be C because we don't know how much CPU write/read respectively consumes; we'll have to monitor to find out...

upvoted 3 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D makes the most sense.

upvoted 1 times

MehulKapadia 1 year, 2 months ago

Selected Answer: D

Keyword: "separate read traffic from write traffic" = Read Replica = Option A and B are not the correct answer.

Option C: Why would you try to have half the resource for read replicas ?. It must be equal resources to ensure read load can be served consistently.

Correct Answer is D: Read replica with same compute power as source database instance.

upvoted 5 times

drich22 1 year ago

By default, a read replica is created with the same storage type as the source DB instance.

By default, a read replica is created with the same storage type as the source DB instance.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html

upvoted 4 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

A: This will not have any change as you are still reading from same instance as you are writing to

B: Not possible (<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>)

C: Why would you do that even if that was possible? No one asked to save on cost

D: Read replicas are normally for handling read-only traffic

upvoted 3 times

ignajtapolandstrong 1 year, 5 months ago

Selected Answer: D

In a Multi-AZ deployment, the standby instance is kept in sync with the primary instance and is used for failover purposes only. You cannot read data from the standby instance in a Multi-AZ deployment. If you need to offload read traffic from the primary instance, you can create one or more Read Replicas. Read Replicas are read-only copies of your database that can be used to offload read traffic from the primary instance, which can help improve performance

upvoted 3 times

Ruffyt 1 year, 7 months ago

D. Configuring the read replicas with the same compute and storage resources as the source database ensures that they can handle the read workload efficiently and provide the required performance boost.

upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Both B and D would work.

Amazon RDS now offers Multi-AZ deployments with readable standby instances (also called Multi-AZ DB cluster deployments). You should consider using Multi-AZ DB cluster deployments with two readable DB instances if you need additional read capacity in your Amazon RDS Multi-AZ deployment and if your application workload has strict transaction latency requirements such as single-digit milliseconds transactions.

<https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/#:~:text=read%20replicas,-Amazon%20RDS,-now%20offers%20Multi>

upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The best solution is to create read replicas for the database and configure them with the same compute and storage resources as the source database.

The key requirements are to quickly optimize performance by isolating reads from writes.

Read replicas allow read-only workloads to be directed to one or more replicas of the source RDS instance. This separates reporting or analytics queries from transactional workloads.

The read replicas should have the same compute and storage as the source to provide equivalent performance for reads. Scaling down the replicas would limit read performance.

Using Multi-AZ alone does not achieve read/write separation. The secondary AZ instance is for disaster recovery, not performance.

upvoted 5 times

MNotABot 1 year, 11 months ago

Read replica + Same resources as we may need to turn replica to primary in few cases

upvoted 1 times

Bmarodi 2 years ago

Selected Answer: D

D meets the requirements.

upvoted 1 times

Adeshina 2 years, 1 month ago

Option C suggests creating read replicas for the database and configuring them with half of the compute and storage resources as the source database. This is a better option as it allows read traffic to be offloaded from the primary database, separating read traffic from write traffic. Configuring the read replicas with half the resources will also save on costs.

upvoted 1 times

NSA_Poker 1 year, 1 month ago

If the source database is already 40% full, the read replica's performance will degrade as it is @ 80% capacity. This will not optimize the apps performance.

upvoted 1 times

Charlesleeee 2 years ago

Err, just curious, what if the production database is 51% full? Your half storage read replica would explode...?

upvoted 5 times

Oldman2023 2 years, 2 months ago

Can anyone explain why B is not an option?

upvoted 5 times

draum010 2 years, 2 months ago

CHATGPT says:

To optimize the application's performance and separate read traffic from write traffic, the solutions architect should recommend creating read replicas for the database and configuring them to serve read requests. Option C and D both suggest creating read replicas, but option D is a better choice because it configures the read replicas with the same compute and storage resources as the source database.

Option A and B suggest changing the existing database to a Multi-AZ deployment, which would provide high availability by replicating the database across multiple Availability Zones. However, it would not separate read and write traffic, so it is not the best solution for optimizing application performance in this scenario.

upvoted 4 times

caffee 2 years, 2 months ago

Multi-AZ: Synchronous replication occurs, meaning that synchronizing data between DB instances immediately can slow down application's performance. But this method increases High Availability.

Read Replicas: Asynchronous replication occurs, meaning that replicating data in other moments rather than in the writing will maintain application's performance. Although the data won't be HA as Multi-AZ kind of deployment, this method increases Scalability. Good for read heavy workloads.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #96

Topic 1

An Amazon EC2 administrator created the following policy associated with an IAM group containing several users:

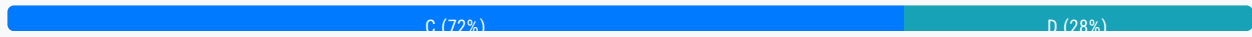
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:TerminateInstances",
      "Resource": "*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "10.100.100.0/24"
        }
      }
    },
    {
      "Effect": "Deny",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "ec2:Region": "us-east-1"
        }
      }
    }
  ]
}
```

What is the effect of this policy?

- A. Users can terminate an EC2 instance in any AWS Region except us-east-1.
- B. Users can terminate an EC2 instance with the IP address 10.100.100.1 in the us-east-1 Region.
- C. Users can terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254. **Most Voted**
- D. Users cannot terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254.

Correct Answer: C

Community vote distribution



Comments

Joxtat Highly Voted 2 years, 5 months ago

What the policy means:

1. Allow termination of any instance if user's source IP address is 100.100.254.
2. Deny termination of instances that are not in the us-east-1 region. Combining this two, you get:
"Allow instance termination in the us-east-1 region if the user's source IP address is 10.100.100.254. Deny termination operation on other regions."

upvoted 73 times

KMohsoe 2 years, 1 month ago

Nice explanation. Thanks

upvoted 4 times

Subh_fidelity Highly Voted 2 years, 6 months ago

C is correct.

0.0/24, the following five IP addresses are reserved:

0.0: Network address.

0.1: Reserved by AWS for the VPC router.

0.2: Reserved by AWS. The IP address of the DNS server is the base of the VPC network range plus two. ...

0.3: Reserved by AWS for future use.

0.255: Network broadcast address.

upvoted 35 times

c12ab95 4 weeks ago

Also, it is the USER's IP that matters, the policy provided does not check or restrict the IP address of the EC2 instance itself

upvoted 1 times

Bmarodi 2 years ago

A good explanation!

upvoted 3 times

Dharmarajan Most Recent 4 months, 2 weeks ago

Selected Answer: C

It is C. Because the keyword "SourceIp". That's what distinguishes C and D.

upvoted 1 times

itsmeDiyan 4 months, 2 weeks ago

Selected Answer: C

Uncle Trump says so

upvoted 1 times

PaulGa 9 months ago

Selected Answer: C

Ans C - must be in us-east-1 region and CIDR address is in allowable range (/24)

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: C

The first rule allows users with the specified IP CIDR to terminate instances, and the second rule specifies that the region must be us-east-1 for the termination process to be allowed, hence C is the correct answer.

upvoted 2 times

iatric 11 months, 2 weeks ago

Selected Answer: C

policy allow us-east-1 and with the specific IP address in the range
upvoted 3 times

ChymKuBoy 12 months ago

Selected Answer: C

C for sure
upvoted 1 times

jhoiti 1 year, 2 months ago

Selected Answer: D

D. Users cannot terminate an EC2 instance in the us-east-1 region when the user's source IP is 10.100.100.254.

This option corresponds to the second statement in the policy, where all EC2 actions in the "us-east-1" region are denied permission when the user's source IP is "10.100.100.254".
upvoted 2 times

lofzee 1 year ago

but it says "StringNotEquals" meaning everything is denied apart from us-east-1
upvoted 4 times

vip2 1 year, 3 months ago

Selected Answer: C

Clearly the answer is C.
D is 'Deny' 'String NOT equal' == only allow us-east-1
upvoted 5 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

Here is how I interpreted this
first part: terminate instance is allowed for the given CIDR block
second part: deny all ec2 actions when region is not us-east-1

so second part is like double negative which means allow for us-east-1 region

You combine both (remember deny always take priority which is why this is written in double negative) and you get:
[allow us-east-region1 to do any action on ec2] when [action is terminate instance and CIDR block is match]
so C is the answer
D is there to confuse you with the double negative
upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: C

Deny takes precedence over Allow. Thus the flow is as follows:

IF region of the EC2 instance is not "us-east-1" -> Deny
ELSE if request is coming from 10.100.100.0/24 -> Allow
ELSE: implicit deny (what is not allowed is denied)
upvoted 4 times

Cyberkayu 1 year, 6 months ago

if
IP = 10.100.100.0/24
allow terminate EC2

Else
Deny EC2 termination permission
- with the condition "String NOT equal" to us-east-1

Answer C

upvoted 5 times

Bjfikky 1 year, 6 months ago

Selected Answer: D

The first statement allows users to terminate EC2 instances (ec2:TerminateInstances) from any IP address within the range 10.100.100.0/24.

The second statement denies users the ability to perform any EC2 actions (ec2:*) in any region other than us-east-1. So, the correct interpretation is:

D. Users cannot terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254

upvoted 1 times

pentium75 1 year, 5 months ago

D denies "the ability to perform any actions in any region OTHER than us-east-1". Thus the user CAN terminate instances IN us-east-1. Thus C.

upvoted 2 times

sweetheatmn 1 year, 7 months ago

Selected Answer: C

C because the explicit deny blocks other regions than us-east-1

upvoted 1 times

tom_cruise 1 year, 8 months ago

Selected Answer: C

The first statement is a subset of the second statement.

upvoted 1 times

prabhjot 1 year, 8 months ago

ans D - This policy denies EC2 instance termination for users with the source IP address 10.100.100.254 in the us-east-1 Region.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #97

Topic 1

A company has a large Microsoft SharePoint deployment running on-premises that requires Microsoft Windows shared file storage. The company wants to migrate this workload to the AWS Cloud and is considering various storage options. The storage solution must be highly available and integrated with Active Directory for access control. Which solution will satisfy these requirements?

- A. Configure Amazon EFS storage and set the Active Directory domain for authentication.
- B. Create an SMB file share on an AWS Storage Gateway file gateway in two Availability Zones.
- C. Create an Amazon S3 bucket and configure Microsoft Windows Server to mount it as a volume.
- D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: D

D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

Amazon FSx for Windows File Server is a fully managed file storage service that is designed to be used with Microsoft Windows workloads. It is integrated with Active Directory for access control and is highly available, as it stores data across multiple availability zones. Additionally, FSx can be used to migrate data from on-premises Microsoft Windows file servers to the AWS Cloud. This makes it a good fit for the requirements described in the question.

upvoted 26 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: D

A. EFS does not provide native integration with AD for access control. While you can configure EFS to work with AD, it requires additional setup and is not as straightforward as using a dedicated Windows file system like FSx for Windows File Server.

B. It may introduce additional complexity for this use case. Creating an SMB file share using AWS Storage Gateway would require maintaining the gateway and managing the synchronization between on-premises and AWS storage.

C. S3 does not natively provide the SMB file protocol required for MS SharePoint and Windows shared file storage. While it is possible to mount an S3 as a volume using 3rd-party tools or configurations, it is not the recommended.

D. FSx for Windows File Server is a fully managed, highly available file storage service that is compatible with MS Windows shared file storage requirements. It provides native integration with AD, allowing for seamless access control and authentication using existing AD user accounts.

upvoted 9 times

Dharmarajan Most Recent 4 months, 2 weeks ago

Selected Answer: D

This Q is simple if we remember - FSx is the only readily windows NTFS/SMB/AD compatible FS in Amazon offering. Please, correct me.

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: D

FSx is most suitable for Windows File servers whereas Active Directory is also important for access control.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans D - Amazon FSx for Windows; set Active Directory

I was initially going for A, but EFS doesn't provide native integration with AD for access control; in any event, its Windows so the obvious answer is FSx

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D for sure

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

"When you create a file system with Amazon FSx, you join it to your Active Directory domain to provide user authentication and file- and folder-level access control."

upvoted 2 times

Ruffyt 1 year, 7 months ago

D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

Amazon FSx for Windows File Server is a fully managed file storage service that is designed to be used with Microsoft Windows workloads. It is integrated with Active Directory for access control and is highly available, as it stores data across multiple availability zones. Additionally, FSx can be used to migrate data from on-premises Microsoft Windows file servers to the AWS Cloud. This makes it a good fit for the requirements described in the question.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

Microsoft Windows shared file storage = Amazon FSx for Windows File Server

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The best solution that satisfies the requirements is D) Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

The key requirements are:

Shared Windows file storage for SharePoint
High availability

high availability
Integrated Active Directory authentication
upvoted 2 times

cheese929 2 years, 1 month ago

Selected Answer: D

D is correct. FSx is for windows and supports AD authentication
upvoted 2 times

kakka22 2 years, 1 month ago

Why not B? Migrating the workload? Maybe is needed a hybrid cloud solution
upvoted 1 times

gx2222 2 years, 2 months ago

Selected Answer: D

One solution that can satisfy the mentioned requirements is to use Amazon FSx for Windows File Server. Amazon FSx is a fully managed service that provides highly available and scalable file storage for Windows-based applications. It is designed to be fully integrated with Active Directory, which allows you to use your existing domain users and groups to control access to your file shares.

Amazon FSx provides the ability to migrate data from on-premises file servers to the cloud, using tools like AWS DataSync, Robocopy or PowerShell. Once the data is migrated, you can continue to use the same tools and processes to manage and access the file shares as you would on-premises.

Amazon FSx also provides features such as automatic backups, data encryption, and native multi-Availability Zone (AZ) deployments for high availability. It can be easily integrated with other AWS services, such as Amazon S3, Amazon EFS, and AWS Backup, for additional functionality and backup options.

upvoted 3 times

psr83 2 years, 5 months ago

Selected Answer: D

FSx is for Windows
upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D
upvoted 1 times

xeun88 2 years, 6 months ago

Im going for D as the answer because FXs is compatible with windows
upvoted 2 times

kajal1206 2 years, 6 months ago

Selected Answer: D

Answer is D
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #98

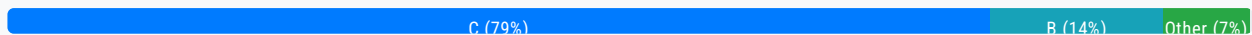
Topic 1

An image-processing company has a web application that users use to upload images. The application uploads the images into an Amazon S3 bucket. The company has set up S3 event notifications to publish the object creation events to an Amazon Simple Queue Service (Amazon SQS) standard queue. The SQS queue serves as the event source for an AWS Lambda function that processes the images and sends the results to users through email.

Users report that they are receiving multiple email messages for every uploaded image. A solutions architect determines that SQS messages are invoking the Lambda function more than once, resulting in multiple email messages.

What should the solutions architect do to resolve this issue with the LEAST operational overhead?

- A. Set up long polling in the SQS queue by increasing the ReceiveMessage wait time to 30 seconds.
- B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.
- C. Increase the visibility timeout in the SQS queue to a value that is greater than the total of the function timeout and the batch window timeout. **Most Voted**
- D. Modify the Lambda function to delete each message from the SQS queue immediately after the message is read before processing.

Correct Answer: C*Community vote distribution***Comments****Six_Fingered_Jose** **Highly Voted** 2 years, 7 months ago**Selected Answer: C**

answer should be C,
users get duplicated messages because -> lambda polls the message, and starts processing the message.
However, before the first lambda can finish processing the message, the visibility timeout runs out on SQS, and SQS returns the message to the poll, causing another Lambda node to process that same message.
By increasing the visibility timeout, it should prevent SQS from returning a message back to the poll before Lambda can finish processing the message
upvoted 67 times

Ello2023 2 years, 4 months ago

I am confused. If the email has been sent many times already why would they need more time?
I believe SQS Queue Fifo will keep in order and any duplicates with same ID will be deleted. Can you tell me where i am going wrong? Thanks
upvoted 9 times

Robrobtutu 2 years, 1 month ago

Increasing the visibility timeout would give time to the lambda function to finish processing the message, which would make it disappear from the queue, and therefore only one email would be send to the user.

If the visibility timeout ends while the lambda function is still processing the message, the message will be returned to the queue and there another lambda function would pick it up and process it again, which would result in the user receiving two or more emails about the same thing.

upvoted 12 times

PaulEkwen 7 months, 2 weeks ago

Increasing the visibility timeout allows a message to remain hidden for a longer period after being received by a Lambda function. However, this only prevents other instances of the function from processing the same message during that time. If the function takes longer to process than expected, messages can still become visible again and be retried, potentially leading to duplicates.

upvoted 3 times

aadityaravi8 1 year, 11 months ago

I agree with your answer explanation

upvoted 1 times

MrAWS 2 years, 4 months ago

I tend to agree with you. See my comments above.

upvoted 2 times

MutiverseAgent 1 year, 11 months ago

I agree it seems solution is C, as thought the SQS FIFO makes sense deduplication id would make NO sense as the system who put messages in the queue is S3 events; and as far as I know S3 do not send duplicated events. Also, the question mention that users are complaining about receiving multiple emails for each email, which is different to say they are receiving occasionally a repeated email; so my guess is SQS FIFO is not needed.

upvoted 2 times

JoeGuan 1 year, 10 months ago

The FIFO SQS is for solving a different problem, where items in the queue require order. You cannot simply switch from a standard queue to fifo queue. Duplicate emails are a common issue with a standard queue. The documentation consistently reminds us that duplicate emails can occur, and the solution is not to create a FIFO queue, but rather adjust the configuration parameters accordingly.

upvoted 5 times

PLN6302 1 year, 9 months ago

amazon s3 doesn't support fifo queues

upvoted 3 times

brushek Highly Voted 2 years, 8 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html>

this is important part:

Immediately after a message is received, it remains in the queue. To prevent other consumers from processing the message again, Amazon SQS sets a visibility timeout, a period of time during which Amazon SQS prevents other consumers from receiving and processing the message. The default visibility timeout for a message is 30 seconds. The minimum is 0 seconds. The maximum is 12 hours.

upvoted 16 times

BennyBrrr Most Recent 4 weeks ago

Selected Answer: C

I guessed B because I know FIFO queues have deduplication functionality, but when I looked it up it wouldn't solve this issue.

FIFO deduplication protects against a producer sending a message with the same deduplication ID to the queue multiple times within the deduplication interval (5 minutes). The question doesn't mention duplicates are being added into the queue; it just mentions that messages from the queue are being processed more than once.

As a result, the answer must be C because messages that take longer to process than the configured visibility timeout are made visible on the queue again, and are therefore picked up by another processor.

upvoted 1 times

AwsAbhiKumar 3 months, 4 weeks ago

Selected Answer: B

I believe answer is B cause FIFO queues provide exactly-once processing and guarantee that messages are delivered in order. They also support message deduplication, which prevents processing duplicate messages within a specified deduplication interval.

Whereas a longer visibility timeout might prevent messages from being processed more than once if processing takes longer than expected. However, it does not eliminate the inherent at-least-once delivery behavior of standard queues, which can still cause duplicates.

upvoted 2 times

jerryI 2 months, 3 weeks ago

I think there is another question that mention the visibility timeout concept as well

My understanding is for SQS, you have producer (inject data to queue) and consumer (get data from queue), the SQS itself does not know who will be the producer or consumer, this is implemented at the producer/consumer side like a Lambda function

So ensuring the read sequence doesnt mean no duplication at all

Instead you need to ensure a data (element in the queue) not to be read multiple times by consumer

And this is done by adjusting the visible timeout, making it long enough will make sure each consumer (Lambda in this case) can either consume the data completely (or fail) without other consumer's interruption

upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: C

SQS feature of "Visibility timeout" sets the time for getting acknowledgement from processor that the message has been successfully processed. I think it should have been called "Invisibility " rather than "Visibility" timeout. Anyways, I think AWS engineers have thought this Queue service through well. So if the processing time > visibility timeout, the message comes back and is picked up by another execution of lambda function. It is such a small thing but it can easily rake up lambda usage if not considered!

upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: B

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html> , standard queue is delivered at least once, the duplication can not be avoided

upvoted 1 times

Rcosmos 5 months ago

Selected Answer: C

C. Aumentar o tempo limite de visibilidade:

Um tempo limite de visibilidade maior reduz a chance de reprocessamento prematuro, mas não resolve completamente o problema de mensagens duplicadas.

É mais complexo ajustar adequadamente o tempo de visibilidade para sincronizar com o processamento da função Lambda e o tempo limite.

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: C

Using Visibility Timeout is the most suitable option here with the least operational overhead.

upvoted 1 times

salman7540 5 months, 3 weeks ago

Selected Answer: C

FIFO queues are designed to never introduce duplicate messages. However, a message producer might introduce duplicates if another producer (lambda in this case) picks up same fifo message after visibility timeout over but previous lambda was still processing it. Hence correct answer is C.

upvoted 1 times

JA2018 6 months, 1 week ago

Selected Answer: B

it should be B

- Using a FIFO (First In, First Out) queue ensures that messages are processed in the exact order they are received, preventing duplicates from being processed multiple times by the Lambda function. By setting up message deduplication IDs, you can further guarantee that even if the same message is sent multiple times, only one instance will be processed.

- When dealing with potential duplicate messages in an event-driven architecture, switching to an SQS FIFO queue with message deduplication is generally the most efficient and reliable solution to prevent redundant processing.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans D - is my preferred choice because it removes the record and therefore eliminates the possibility of duplication; the only snag here is ensuring the SQS record is locked.

Ans C, which has the most votes will work but is less robust - but perhaps it is the cheapest option.

Ans A, the author's preference, won't work unless the Lambda function is guaranteed to complete within 30secs - and nowhere is that stated

upvoted 2 times

jaradat02 10 months, 3 weeks ago

the correct answer is C, although B and D might also work, C is the one that achieves the least operational overhead condition.

upvoted 1 times

Solomon2001 1 year, 1 month ago

Selected Answer: B

B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.

Here's why this option is suitable:

SQS FIFO Queue:

Amazon SQS FIFO (First-In-First-Out) queues ensure that the order of messages is preserved.

Each message in a FIFO queue has a unique message deduplication ID.

By using a FIFO queue, you can prevent duplicate messages from being processed.

Message Deduplication ID:

When sending messages to a FIFO queue, set the message deduplication ID to ensure that identical messages are treated as duplicates.

If a message with the same deduplication ID is sent within a 5-minute window, it is considered a duplicate and discarded.

<https://docs.aws.amazon.com/lambda/latest/dg/with-sqs.html>

upvoted 2 times

Anthony_Rodrigues 1 year ago

This wouldn't work as expected because the issue is not the order or the queue receiving multiple times the same message. The issue is that the Lambda executes the same message multiple times.

The FIFO deduplication ID works when the producer sends the message, not when the consumer is receiving the message. (<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/using-messagededuplicationid-property.html>)

For this to work, the Lambda function would need to know/keep the last N deduplication ID to guarantee that it isn't processing the same message, which goes against Lambda's statelessness.

Therefore, for this case, B is the best answer.

upvoted 1 times

824c449 1 year, 1 month ago

Selected Answer: B

B: Directly addresses the issue of duplicates by ensuring exact-once processing and message ordering, which mitigates the risk without requiring adjustments based on anticipated processing times or additional application logic to manage potential

...which requires adjustments based on anticipated processing times or additional application logic to manage potential duplicates.

So, while increasing the visibility timeout could help in managing when messages are available for processing again, it doesn't provide a structural solution to the problem of duplicate processing in the way that using an SQS FIFO queue does, nor does it ensure the operational simplicity and reliability that comes with eliminating duplicates at the source.

upvoted 2 times

ManikRoy 1 year, 1 month ago

Selected Answer: C

to all those who are considering option B, Its INCORRECT.

Amazon Simple Queue Service FIFO (First-In-First-Out) queues aren't supported as an Amazon S3 event notification destination. <https://docs.aws.amazon.com/AmazonS3/latest/userguide/EventNotifications.html>

upvoted 3 times

Uzbekistan 1 year, 2 months ago

Selected Answer: B

Given the requirement to resolve the issue of multiple email messages being sent to users with the least operational overhead, the most appropriate solution is:

B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.

Explanation:

SQS FIFO Queue: FIFO (First-In-First-Out) queues in SQS ensure that the order in which messages are sent and received is strictly preserved and that each message is processed only once. By switching to an SQS FIFO queue, you can prevent the Lambda function from processing duplicate messages.

upvoted 1 times

ManikRoy 1 year, 1 month ago

Its INCORRECT

Amazon Simple Queue Service FIFO (First-In-First-Out) queues aren't supported as an Amazon S3 event notification destination.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/EventNotifications.html>

upvoted 1 times

vip2 1 year, 3 months ago

Selected Answer: C

'Visibility Timeout' is suitable(better) solution to solve the issue.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #99

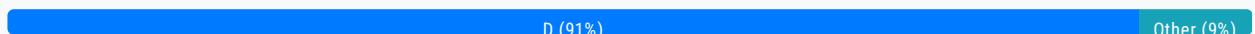
Topic 1

A company is implementing a shared storage solution for a gaming application that is hosted in an on-premises data center. The company needs the ability to use Lustre clients to access data. The solution must be fully managed. Which solution meets these requirements?

- A. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.
- B. Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
- C. Create an Amazon Elastic File System (Amazon EFS) file system, and configure it to support Lustre. Attach the file system to the origin server. Connect the application server to the file system.
- D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

123jhl0 **Highly Voted** 2 years, 7 months ago

Selected Answer: D

Answer is D.
Lustre in the question is only available as FSx
<https://aws.amazon.com/fsx/lustre/>
upvoted 30 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: D

Option D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

Amazon FSx for Lustre is a fully managed file system that is designed for high-performance workloads, such as gaming applications. It provides a high-performance, scalable, and fully managed file system that is optimized for Lustre clients, and it is fully integrated with Amazon EC2. It is the only option that meets the requirements of being fully managed and able to

support Lustre clients.

upvoted 16 times

Dharmarajan Most Recent 4 months, 2 weeks ago

Selected Answer: D

This one had me stumped! Thanks, community, for the link.

upvoted 1 times

Rcosmos 5 months ago

Selected Answer: C

Não é uma prática recomendada:

O padrão em sistemas assíncronos com SQS e Lambda é remover a mensagem apenas após o processamento bem-sucedido. Isso garante que, caso algo falhe, a mensagem possa ser processada novamente.

Não resolve duplicidade real:

Mesmo que você exclua a mensagem imediatamente, o problema de mensagens duplicadas pode ocorrer devido à lógica de envio duplicado no S3 ou por notificações redundantes do evento de criação de objetos. Assim, o problema original pode persistir.

upvoted 1 times

Rcosmos 5 months ago

Selected Answer: C

A opção D pode parecer uma solução direta, mas não é recomendada porque introduz riscos significativos para a confiabilidade do sistema. Vamos analisar mais a fundo:

O que a opção D sugere:

Modificar a função Lambda para excluir cada mensagem da fila do SQS imediatamente após a mensagem ser lida, antes do processamento.

Problemas com a opção D:

Perda de mensagens em caso de falhas:

Se a mensagem for excluída do SQS antes de ser processada e algo der errado (como falha na função Lambda ou erro na lógica de processamento), a mensagem será perdida permanentemente.

O SQS é projetado para garantir que a mensagem seja processada com segurança antes de ser removida. Excluir a mensagem antes do processamento viola essa garantia.

upvoted 1 times

satyaammm 5 months, 1 week ago

Selected Answer: D

Amazon FSx Lustre is the most suitable for managing Lustre Clients.

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D, self explanatory.

upvoted 2 times

lofzee 1 year ago

the question is missing some text i think... as none of the answers really solve the equation.

if the answer is D, how is the on prem infrastructure accessing Lustre in AWS?

either way the only possible option it could be is D, so D it is.

upvoted 3 times

NullM 1 year, 1 month ago

Sorry but what is origin server in this context, does it mean AWS server or premise??

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

D: Lustre is key requirement
AB: No support for Lustre
C: Cannot just configure EFS to support Lustre file system
upvoted 2 times

Ruffyit 1 year, 7 months ago

Option D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

Amazon FSx for Lustre is a fully managed file system that is designed for high-performance workloads, such as gaming applications. It provides a high-performance, scalable, and fully managed file system that is optimized for Lustre clients, and it is fully integrated with Amazon EC2. It is the only option that meets the requirements of being fully managed and able to support Lustre clients.

upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The correct solution is D) Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

The key requirements are:

Shared storage solution

Support Lustre clients

Fully managed service

Amazon FSx for Lustre provides a fully managed file system that is optimized for Lustre workloads. It allows Lustre clients to seamlessly connect to the file system.

upvoted 4 times

RupeC 1 year, 10 months ago

Selected Answer: A

Sorry, but I disagree with everyone. The question states "a gaming application that is hosted in an on-premises data center". Option D does not address this and cannot to my knowledge address it. Thus:

A. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.

By using AWS Storage Gateway in file gateway mode, you can extend your on-premises data center storage into the AWS cloud. The file share created on AWS Storage Gateway can use the necessary client protocol (such as Lustre), which would allow the Lustre clients in your on-premises data center to access the data stored on AWS Storage Gateway.

This solution enables you to use Lustre clients to access data, while still keeping the gaming application hosted in your on-premises data center. AWS Storage Gateway provides a fully managed solution for this hybrid scenario, allowing seamless integration between on-premises and AWS cloud storage.

upvoted 5 times

David_Ang 1 year, 8 months ago

mate if you have an aws service that is meant to be used for this task, there is simply not discussion, is more simple, is more cheap and better option

upvoted 1 times

JoeGuan 1 year, 10 months ago

So, I think that the FSx File Gateway is currently only available for Windows? I don't think Lustre is part of this offering yet as of 8/8/2023

upvoted 1 times

pentium75 1 year, 5 months ago

"The file share created on AWS Storage Gateway can use the necessary client protocol (such as Lustre)", no it can use only NFS
AWS Storage Gateway can use the necessary client protocol (such as Lustre), no it can use only NFS

or SMB. But NOT Lustre.

upvoted 3 times

james2033 1 year, 11 months ago

Selected Answer: D

Content of "Amazon FSx for Lustre" at this link <https://aws.amazon.com/fsx/lustre/> . Focus at image, section: "On-premises clients".

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: D

A. Lustre client access is not supported by AWS Storage Gateway file gateway.

B. Creating a Windows file share on an EC2 Windows instance is suitable for Windows-based file sharing, but it does not provide the required Lustre client access. Lustre is a high-performance parallel file system primarily used in high-performance computing (HPC) environments.

C. EFS does not natively support Lustre client access. Although EFS is a managed file storage service, it is designed for general-purpose file storage and is not optimized for Lustre workloads.

D. Amazon FSx for Lustre is a fully managed file system optimized for high-performance computing workloads, including Lustre clients. It provides the ability to use Lustre clients to access data in a managed and scalable manner. By choosing this option, the company can benefit from the performance and manageability of Amazon FSx for Lustre while meeting the requirement of Lustre client access.

upvoted 4 times

Musti35 2 years, 1 month ago

Selected Answer: D

https://aws.amazon.com/fsx/lustre/?nc1=h_ls#:~:text=Amazon%20FSx%20for%20Lustre%20provides%20fully%20managed%20shared%20storage%20with%20the%20scalability%20and%20performance%20of%20the%20popular%20Lustre%20file%20system.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

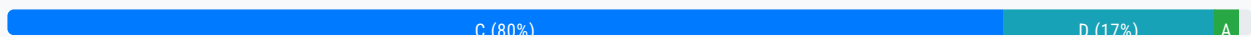
Question #100

Topic 1

A company's containerized application runs on an Amazon EC2 instance. The application needs to download security certificates before it can communicate with other business applications. The company wants a highly secure solution to encrypt and decrypt the certificates in near real time. The solution also needs to store data in highly available storage after the data is encrypted.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create AWS Secrets Manager secrets for encrypted certificates. Manually update the certificates as needed. Control access to the data by using fine-grained IAM access.
- B. Create an AWS Lambda function that uses the Python cryptography library to receive and perform encryption operations. Store the function in an Amazon S3 bucket.
- C. Create an AWS Key Management Service (AWS KMS) customer managed key. Allow the EC2 role to use the KMS key for encryption operations. Store the encrypted data on Amazon S3. **Most Voted**
- D. Create an AWS Key Management Service (AWS KMS) customer managed key. Allow the EC2 role to use the KMS key for encryption operations. Store the encrypted data on Amazon Elastic Block Store (Amazon EBS) volumes.

Correct Answer: C*Community vote distribution***Comments****Chunslil** **Highly Voted** 2 years, 8 months ago

C makes a better sense. Between C (S3) and D (EBS), S3 is highly available with LEAST operational overhead.
upvoted 50 times

MutiverseAgent 1 year, 11 months ago

Agree, also the data in EBS will be accessible only to the EC2 instance and that is not as available as S3 would be.
upvoted 8 times

jerryl 2 months, 3 weeks ago

agree on this, although the question did not mention where the EBS is attached to, but i assume should be the EC2, then your statement make sense

upvoted 1 times

MXB05 Highly Voted 2 years, 8 months ago

Selected Answer: C

Correct Answer is C: EBS is not highly available

upvoted 23 times

TS1991 8 months, 2 weeks ago

EBS volumes are not Multi-AZ. EBS io2 types are multi-attach within the same AZ. EFS is multi-AZ

upvoted 2 times

FNJ1111 2 years, 5 months ago

Per AWS: "Amazon EBS volumes are designed to be highly available, reliable, and durable"

<https://aws.amazon.com/ebs/features/>

upvoted 2 times

Ello2023 2 years, 4 months ago

EBS is Highly Available as it stores in multi AZ and S3 is regional.

upvoted 2 times

oguz11 2 years, 4 months ago

EBS also has Multi-AZ capability, but it does not replicate the data across multiple availability zones by default. When Multi-AZ is enabled, it creates a replica of the EBS volume in a different availability zone and automatically failover to the replica in case of a failure. However, this requires additional configuration and management. In comparison, Amazon S3 automatically replicates data across multiple availability zones without any additional configuration. Therefore, storing the data on Amazon S3 provides a simpler and more efficient solution for high availability.

upvoted 11 times

dkw2342 1 year, 3 months ago

This is false. There is no AWS-provided functionality that will replicate EBS volumes across AZs. There are 3rd-party solutions to this, but that's not what's being asked here.

EBS is only replicated WITHIN an AZ by default.

upvoted 3 times

Bayebrymo 1 year, 1 month ago

This is false... S3 is Multi AZ and EBS is only replicated WITHIN an AZ by default.

upvoted 3 times

pentium75 1 year, 5 months ago

S3 is also highly available. Within the region, but still. Multi-AZ = HA.

upvoted 3 times

JayBee65 2 years, 5 months ago

Yes it is!

upvoted 1 times

pabloveintimilla Most Recent 3 months, 3 weeks ago

Selected Answer: C

C. Correct, AWS KMS handles encryption with minimal effort and Amazon S3 offers durable and available storage vs D Incorrect. EBS is storage attached to an EC2 instance, which reduces availability compared to S3.

upvoted 1 times

sk1974 3 months, 4 weeks ago

Selected Answer: C

The answer is in LEAST operational overhead

upvoted 1 times

upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: C

Of the given options, C makes most sense. Reason being the rest of the options do not make as much sense due to A. Being not specific enough, B. being insufficient to achieve the objective, D, being on EBS, which needs to be attached to a EC2 instance.

One thing with the questions is that there is many times, some data that is unclear or there is some ambiguity. I feel these scenarios makes one to assume things and perhaps even train the mind to evaluate ambiguous situations. This is valuable in my opinion.

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: C

S3 is highly available compared to EBS and using AWS KMS is more suitable for managing certificates here

upvoted 1 times

thiahthura 6 months, 1 week ago

Selected Answer: C

Both C and D are correct. C is suitable for this requirement. we've to use S3 because they want to save the data with H.A

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: C

C is the most efficient.

upvoted 1 times

Shub80 11 months, 3 weeks ago

AWS KMS: Provides a managed service for secure key storage and encryption/decryption operations. This eliminates the need to manage encryption/decryption logic within the application itself.

Customer Managed Key: The company maintains control over the key, ensuring security.

EC2 Role Permissions: Granting permissions to the EC2 role allows the application to use KMS for encryption/decryption without managing individual credentials.

Amazon S3: Offers highly available and scalable storage for the encrypted certificates. S3 is generally cheaper than EBS for data that is not frequently accessed.

upvoted 4 times

ChymKuBoy 12 months ago

Selected Answer: C

C for sure

upvoted 1 times

huangyou2003 1 year, 1 month ago

Selected Answer: C

S3: highly available

EBS: lower latency

upvoted 1 times

f761d0e 1 year, 1 month ago

"Amazon S3 is an object storage service that can store large volumes of unstructured data, whereas Amazon EBS is a block storage service that is ideally suited for durable, low-latency data storage associated with EC2 instances."

<https://www.tutorialspoint.com/difference-between-amazon-s3-and-amazon-ebs#:~:text=In%20conclusion%2C%20Amazon%20S3%20is,storage%20associated%20with%20EC2%20instances.>

Seems like D to me. S3 is for large data, EBS is ec2 specific.

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

The language is confusing over here so I'm going by process of elimination

A: Wrong because manual operation and fine grained IAM is overhead

D: Wrong

b: wnatr

D: Between C and D S3 is more HA than EFS so C wins

upvoted 2 times

awsgeek75 1 year, 5 months ago

Sorry meant EBS, not EFS for D

D: Between C and D, S3 is more HA than EBS. So C wins

upvoted 2 times

ignajtpolandstrong 1 year, 5 months ago

Selected Answer: D

I would select D.

you can mount a single Amazon Elastic Block Store (EBS) volume to multiple Docker containers running on the same Amazon Elastic Compute Cloud (EC2) instance.

you can store data from a container running on Amazon Elastic Compute Cloud (EC2) to an Amazon Simple Storage Service (S3) bucket. One way to do this is to use the `aws s3 cp` command in the command line of the EC2 instance.

upvoted 1 times

pentium75 1 year, 5 months ago

Selected Answer: C

A - does not mention storing the encrypted data at all (though that is a requirement), also involves manual action which is surely NOT "least operational effort"

B - Doesn't make any sense

C - Yes, S3 meets the requirements and is easy to access from containerized app

D - EBS volumes are mounted to the container host, but data is created on containers

upvoted 3 times

xdkonorek2 1 year, 7 months ago

Selected Answer: A

A is OK

secrets manager:

- is highly available

- you can store custom secrets in it like certificate

- automatically encrypts secrets at rest, and can be configured for encryption in transit

- downloading certificate from it is less operational overhead than decrypting it manually with KMS key

arguments against it that this is more manual than C and D? this manual step is necessary measure and can't be omitted in other options

C and D have this "store the encrypted data in..." to store encrypted certificate you have to: log in to instance, get kms key, get certificate, encrypt it, and load that data this is more operational overhead

upvoted 2 times

pentium75 1 year, 5 months ago

"Least operational overhead" and "manually" (as in A) usually don't go together. Also, A does not say anything about storing the data (which is a requirement).

"C and D have this 'store the encrypted data in'" yes, exactly, the encrypted data, NOT the certificate. You encrypt data with the certificate, and you want to store THAT encrypted data.

upvoted 3 times

David_Ang 1 year, 8 months ago

Selected Answer: C

"C" is more correct because S3 is more efficient and cheaper to store data like certificates, like this case. Also Option D involves using Amazon Elastic Block Store (Amazon EBS) volumes, which is not typically used for storing certificates and may introduce unnecessary complexity and operational overhead.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #101

Topic 1

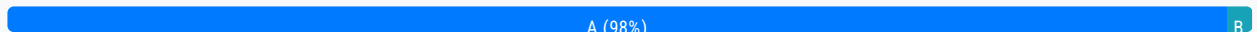
A solutions architect is designing a VPC with public and private subnets. The VPC and subnets use IPv4 CIDR blocks. There is one public subnet and one private subnet in each of three Availability Zones (AZs) for high availability. An internet gateway is used to provide internet access for the public subnets. The private subnets require access to the internet to allow Amazon EC2 instances to download software updates.

What should the solutions architect do to enable Internet access for the private subnets?

- A. Create three NAT gateways, one for each public subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT gateway in its AZ. **Most Voted**
- B. Create three NAT instances, one for each private subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT instance in its AZ.
- C. Create a second internet gateway on one of the private subnets. Update the route table for the private subnets that forward non-VPC traffic to the private internet gateway.
- D. Create an egress-only internet gateway on one of the public subnets. Update the route table for the private subnets that forward non-VPC traffic to the egress-only Internet gateway.

Correct Answer: A

Community vote distribution



Comments

Gil80 **Highly Voted** 2 years, 7 months ago

Selected Answer: A

NAT Instances - OUTDATED BUT CAN STILL APPEAR IN THE EXAM!
However, given that A provides the newer option of NAT Gateway, then A is the correct answer.

B would be correct if NAT Gateway wasn't an option.
upvoted 19 times

Shrestwt 2 years, 1 month ago

NAT instance or NAT Gateway always created in public subnet to provide internet access to private subnet. In option B. they are creating NAT Instance in private subnet which is not correct.
upvoted 29 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: A

The correct answer is option A.

To enable Internet access for the private subnets, the solutions architect should create three NAT gateways, one for each public subnet in each Availability Zone (AZ). NAT gateways allow private instances to initiate outbound traffic to the Internet but do not allow inbound traffic from the Internet to reach the private instances.

The solutions architect should then create a private route table for each AZ that forwards non-VPC traffic to the NAT gateway in its AZ. This will allow instances in the private subnets to access the Internet through the NAT gateways in the public subnets.

upvoted 12 times

SirDNS Most Recent 2 months, 3 weeks ago

Selected Answer: A

NAT Gateways always have to part of public subnet. We can then create a destination to these NAT Gateways in the private route table.

upvoted 1 times

satyaamm 5 months, 1 week ago

Selected Answer: A

NAT Gateways are much better here as they can easily scale and provide high availability. NAT instances cant provide scalability and HA. Internet Gateway is 2-way and not suitable here. Egress-only is used for IPv6.

upvoted 1 times

kernel1973 6 months, 3 weeks ago

You have to create 3 NAT gateways but in the private subnet.

Public and Private Subnets are the name of the concept . A public subnet is a subnet with a route to the internet gateway , private subnet doesn't have a route to the internet gateway.

In this case the private subnets must have the NAT Gateway/NAT Instances , not the public subnet because if we are defined a subnet as public it means that this subnet has a route to the IG.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: A

Ans A - it can only be A or B and NAT Gateways are preferred over NAT Instances.

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

the correct answer is A, to connect a private subnet to the internet using internet gateways is irrelevant, you have to use either NAT gateway or NAT instance, and NAT gateway is the better choice.

upvoted 1 times

soufiyane 1 year, 2 months ago

Selected Answer: A

Nat instances can do the same except it's not cost effective also it need a lot of managment, going with nat gateways makes more sense

upvoted 1 times

ronin201 1 year, 7 months ago

in Azure there is 1 NAT GW multi AZ, 1 per network, I think this is example for AWS to change

upvoted 1 times

pentium75 1 year, 5 months ago

But in AWS a NAT GW is attached to a subnet, and a subnet resides in a single AZ. Can't create multi-AZ NAT GW without changing whole architecture. You CAN use one NAT GW from multiple subnets in multiple AZs I think, but then it would not be HA.

upvoted 1 times

Ruffyt 1 year, 7 months ago

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-example-private-subnets-nat.html>

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

The best solution is to create a NAT gateway in each public subnet (one per availability zone), and update the route tables for the private subnets to send internet traffic to the NAT gateway.

NAT gateways allow private subnets to access the internet for things like software updates, without exposing those instances directly to the internet. An egress-only internet gateway would allow outbound access, but also allow inbound internet traffic, which is not desired for the private subnets.

upvoted 5 times

james2033 1 year, 10 months ago

Selected Answer: A

"Egress" means outbound connection, remove D. "Second gateway", remove C.

Now has only A and B. The different between A versus B is "1 NAT gateway, 1 for public subnet in each AZ" (A) and "1 NAT gateway, 1 for private subnet in each AZ" (B).

Choose A.

upvoted 4 times

cookieMr 1 year, 11 months ago

By creating a NAT gateway in each public subnet, the private subnets can route their Internet-bound traffic through the NAT gateways. This allows EC2 in the private subnets to download software updates and access other resources on the Internet.

Additionally, a separate private route table should be created for each AZ. The private route tables should have a default route that forwards non-VPC traffic (0.0.0.0/0) to the corresponding NAT gateway in the same AZ. This ensures that the private subnets use the appropriate NAT gateway for Internet access.

B is incorrect because NAT instances require manual management and configuration compared to NAT gateways, which are a fully managed service. NAT instances are also being deprecated in favor of NAT gateways.

C is incorrect because creating a second internet gateway on a private subnet is not a valid solution. Internet gateways are associated with public subnets and cannot be directly associated with private subnets.

D is incorrect because egress-only internet gateways are used for IPv6 traffic.

upvoted 5 times

Jeeva28 2 years ago

NAT Gateway will be created Public Subnet and Provide access to Private Subnet

upvoted 1 times

cheese929 2 years, 1 month ago

Selected Answer: A

A is correct.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-example-private-subnets-nat.html>

upvoted 2 times

Heric 2 years, 1 month ago

Selected Answer: A

Now NAT Instances is avoided by AWS. Then choose the NAT Gateway

upvoted 3 times

alexiscloud 2 years, 2 months ago

A: NAT Gateway

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #102

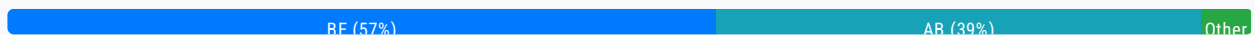
Topic 1

A company wants to migrate an on-premises data center to AWS. The data center hosts an SFTP server that stores its data on an NFS-based file system. The server holds 200 GB of data that needs to be transferred. The server must be hosted on an Amazon EC2 instance that uses an Amazon Elastic File System (Amazon EFS) file system. Which combination of steps should a solutions architect take to automate this task? (Choose two.)

- A. Launch the EC2 instance into the same Availability Zone as the EFS file system.
- B. Install an AWS DataSync agent in the on-premises data center. **Most Voted**
- C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.
- D. Manually use an operating system copy command to push the data to the EC2 instance.
- E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. **Most Voted**

Correct Answer: BE

Community vote distribution



Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: BE

Answer and HOW-TO

- B. Install an AWS DataSync agent in the on-premises data center.
- E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server.

To automate the process of transferring the data from the on-premises SFTP server to an EC2 instance with an EFS file system, you can use AWS DataSync. AWS DataSync is a fully managed data transfer service that simplifies, automates, and accelerates transferring data between on-premises storage systems and Amazon S3, Amazon EFS, or Amazon FSx for Windows File Server.

To use AWS DataSync for this task, you should first install an AWS DataSync agent in the on-premises data center. This agent is a lightweight software application that you install on your on-premises data source. The agent communicates with the AWS DataSync service to transfer data between the data source and target locations.

upvoted 72 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Next, you should use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. A location

next, you should use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. A location represents a data source or a data destination in an AWS DataSync task. You can create a location for the on-premises SFTP server by specifying the IP address, the path to the data, and the necessary credentials to access the data.

Once you have created the location configuration for the on-premises SFTP server, you can use AWS DataSync to transfer the data to the EC2 instance with the EFS file system. AWS DataSync handles the data transfer process automatically and efficiently, transferring the data at high speeds and minimizing downtime.

upvoted 23 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Explanation of other options

A. Launch the EC2 instance into the same Availability Zone as the EFS file system.

This option is not wrong, but it is not directly related to automating the process of transferring the data from the on-premises SFTP server to the EC2 instance with the EFS file system. Launching the EC2 instance into the same Availability Zone as the EFS file system can improve the performance and reliability of the file system, as it reduces the latency between the EC2 instance and the file system. However, it is not necessary for automating the data transfer process.

upvoted 17 times

Buruguduystunstugudunstuy 2 years, 5 months ago

C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.

This option is incorrect because Amazon EBS is a block-level storage service that is designed for use with Amazon EC2 instances. It is not suitable for storing large amounts of data that need to be accessed by multiple EC2 instances, like in the case of the NFS-based file system on the on-premises SFTP server. Instead, you should use Amazon EFS, which is a fully managed, scalable, and distributed file system that can be accessed by multiple EC2 instances concurrently.

upvoted 9 times

Buruguduystunstugudunstuy 2 years, 5 months ago

D. Manually use an operating system copy command to push the data to the EC2 instance.

This option is not wrong, but it is not the most efficient or automated way to transfer the data from the on-premises SFTP server to the EC2 instance with the EFS file system. Manually transferring the data using an operating system copy command would require manual intervention and would not scale well for large amounts of data. It would also not provide the same level of performance and reliability as a fully managed service like AWS DataSync.

upvoted 8 times

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: AB

****A****. Launch the EC2 instance into the same Availability Zone as the EFS file system.

Makes sense to have the instance in the same AZ the EFS storage is.

****B****. Install an AWS DataSync agent in the on-premises data center.

The DataSync will move the data to the EFS, which already uses the EC2 instance (see the info provided). No more things are required...

C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.

This secondary EBS volume isn't required... the data should be moved on to EFS...

D. Manually use an operating system copy command to push the data to the EC2 instance.

Potentially possible (instead of A), BUT the "automate this task" premise goes against any "manually" action. So, we should keep A.

E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server.

I don't get the relationship between DataSync and the configuration for SFTP "on-prem"! Nonsense.

So, answers are A&B

upvoted 61 times

Lalo 2 years, 3 months ago

CORRECT ANSWER: B&E

Steps 4 & 5

https://aws.amazon.com/datasync/getting-started/?nc1=h_ls

upvoted 16 times

Iconique 1 year, 8 months ago

Just go to AWS Console, to DataSync and choose "Create Location Configuration". Locations configurations are endpoints used in DataSync task. A location can be the source endpoint of the task, e.g. a NFS on-premise filesystem. So E is helping in

the automation process. A is not even part of this automation process, it is a solution already agreed to have EC2 with EFS, how you connect EC2 to EFS is not part of the solution!

upvoted 6 times

RBSK 2 years, 6 months ago

will A,B work without E?

upvoted 3 times

Cizzla7049 2 years, 6 months ago

E is correct

<https://aws.amazon.com/blogs/storage/migrating-storage-with-aws-datasync/>

upvoted 4 times

happieeee 1 year, 3 months ago

Use AWS Transfer Family instead of DataSync for SFTP. So E seems incorrect.

When do I use AWS DataSync and when do I use AWS Transfer Family?

A: If you currently use SFTP to exchange data with third parties, AWS Transfer Family provides a fully managed SFTP, FTPS, FTP, and AS2 transfer directly into and out of Amazon S3, while reducing your operational burden.

If you want an accelerated and automated data transfer between NFS servers, SMB file shares, Hadoop clusters, self-managed or cloud object storage, AWS Snowcone, Amazon S3, Amazon EFS, and Amazon FSx, you can use AWS DataSync. DataSync is ideal for customers who need online migrations for active data sets, timely transfers for continuously generated data, or replication for business continuity.

upvoted 1 times

Vandaman Most Recent 3 months, 3 weeks ago

Selected Answer: BE

I was initially going with AB, but I questioned why would you need to launch an EC2 Instance for?

upvoted 2 times

satyaamm 5 months, 1 week ago

Selected Answer: BE

Since DataSync needs to be installed locally on on-premises and also we need DataSync to use SFTP.

upvoted 2 times

Gizmo2022 5 months, 4 weeks ago

I'm going to go with AB because AWS DataSync does not support SFTP location configuration.

upvoted 1 times

tom_cruise 6 months, 2 weeks ago

Selected Answer: BE

A is wrong because EFS is across AZ.

upvoted 1 times

Ode7d1b 6 months, 3 weeks ago

Selected Answer: AB

A. Launch the EC2 instance into the same Availability Zone as the EFS file system:

Amazon EFS file systems are tied to specific Availability Zones (AZs) within a region, and for optimal performance and availability, the EC2 instance should be launched in the same AZ as the EFS file system. This ensures low-latency access to the EFS data.

B. Install an AWS DataSync agent in the on-premises data center:

AWS DataSync is a service that automates data transfer between on-premises storage and AWS. To transfer the data from the on-premises server to EFS, you would need to install the DataSync agent on your on-premises system. The agent securely handles data transfer, ensuring the process is efficient and reliable.

upvoted 2 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: AB

A, B for sure
upvoted 1 times

PaulGa 9 months ago

Selected Answer: AB

Ans A,B - hint: "The server must be hosted on an Amazon EC2 instance that uses an Amazon Elastic File System (Amazon EFS) file system" then use DataSync on-prem
upvoted 2 times

XXXXXINN 9 months, 1 week ago

option E is not valid because AWS DataSync does not support SFTP as a location configuration. Instead, you would need to use AWS Transfer Family for SFTP transfers.
upvoted 3 times

SaurabhTiwari1 9 months, 2 weeks ago

Selected Answer: BE

BE is correct
upvoted 1 times

KTEgghead 10 months, 2 weeks ago

Selected Answer: BE

let DataSync work it all out - AWS DataSync can create a suitable location configuration for an on-premises SFTP server.
upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: BE

B and E is the most logical solution, launching the instance in the same AZ as the EFS is not crucial, C and D negate the automation part and the part where it says that we need to use EFS.
upvoted 2 times

freedafeng 11 months, 1 week ago

I don't think E is correct. You create an EC2, and DataSync to migrate the NFS to EFS. That's it. You don't need to migrate anything on the on-premise sftp server
upvoted 1 times

jatric 11 months, 2 weeks ago

Selected Answer: BE

A could be the right choice but here have to choose 2 options so alone with AB and with E it won't work so correct combination is BE
upvoted 1 times

ChymKuBoy 12 months ago

Selected Answer: BE

BE for sure
upvoted 1 times

lofzee 1 year ago

did think AB and now im thinking B and E. the questions and answers seem a bit vague like something is missing but have a look at the info on the AWS page. Use Datasync for migrations between NFS servers.

Q: When do I use AWS DataSync and when do I use AWS Transfer Family?

A: If you currently use SFTP to exchange data with third parties, AWS Transfer Family provides a fully managed SFTP, FTPS, FTP, and AS2 transfer directly into and out of Amazon S3, while reducing your operational burden.

If you want an accelerated and automated data transfer between NFS servers, SMB file shares, Hadoop clusters, self-managed or cloud object storage, AWS Snowcone, Amazon S3, Amazon EFS, and Amazon FSx, you can use AWS DataSync. DataSync is ideal for customers who need online migrations for active data sets, timely transfers for continuously generated data, or

replication for business continuity.
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #103

Topic 1

A company has an AWS Glue extract, transform, and load (ETL) job that runs every day at the same time. The job processes XML data that is in an Amazon S3 bucket. New data is added to the S3 bucket every day. A solutions architect notices that AWS Glue is processing all the data during each run.

What should the solutions architect do to prevent AWS Glue from reprocessing old data?

- A. Edit the job to use job bookmarks. **Most Voted**
- B. Edit the job to delete data after the data is processed.
- C. Edit the job by setting the NumberOfWorkers field to 1.
- D. Use a FindMatches machine learning (ML) transform.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jh10 **Highly Voted** 2 years, 1 month ago

Selected Answer: A

This is the purpose of bookmarks: "AWS Glue tracks data that has already been processed during a previous run of an ETL job by persisting state information from the job run. This persisted state information is called a job bookmark. Job bookmarks help AWS Glue maintain state information and prevent the reprocessing of old data."

<https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html>

upvoted 51 times

cookieMr **Highly Voted** 1 year, 5 months ago

Selected Answer: A

A. Job bookmarks in Glue allow you to track the last-processed data in a job. By enabling job bookmarks, Glue keeps track of the processed data and automatically resumes processing from where it left off in subsequent job runs.

B. Results in the permanent removal of the data from the S3, making it unavailable for future job runs. This is not desirable if the data needs to be retained or used for subsequent analysis.

C. It would only affect the parallelism of the job but would not address the issue of reprocessing old data. It does not provide a mechanism to track the processed data or skip already processed data.

D. It is not directly related to preventing Glue from reprocessing old data. The FindMatches transform is used for identifying

D: It is not directly related to preventing Glue from reprocessing old data. The FindMatches transform is used for identifying and matching duplicate or matching records in a dataset. While it can be used in data processing pipelines, it does not address the specific requirement of avoiding reprocessing old data in this scenario.

upvoted 12 times

awsgeek75 Most Recent 11 months ago

Selected Answer: A

B: Glue can delete DataSet but this option is too vague to consider or too open to mean anything

C: Won't help with repeated ETL. This property affects parallelism

D: Too vague

upvoted 1 times

Ruffyit 1 year, 1 month ago

<https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html>

upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: A

The best solution is to edit the AWS Glue job to use job bookmarks.

Job bookmarks allow AWS Glue ETL jobs to track which data has already been processed during previous runs. This prevents reprocessing of old data.

Deleting the data after processing would cause the data to be lost and unavailable for future processing. Reducing the number of workers may improve performance but does not prevent reprocessing of old data. Using a FindMatches ML transform is used for record matching, not preventing reprocessing.

So the solutions architect should enable job bookmarks in the AWS Glue job configuration. This will allow the ETL job to keep track of processed data and only transform the new data added since the last run.

upvoted 1 times

bedwal2020 1 year, 7 months ago

Selected Answer: A

Job bookmark to make sure that the glue job will not process already processed files.

upvoted 1 times

Heric 1 year, 7 months ago

Selected Answer: A

Job bookmarks are used in AWS Glue ETL jobs to keep track of the data that has already been processed in a previous job run. With bookmarks enabled, AWS Glue will read the bookmark information from the previous job run and will only process the new data that has been added to the data source since the last job run. This saves time and reduces costs by eliminating the need to reprocess old data.

Therefore, a solutions architect should edit the AWS Glue ETL job to use job bookmarks so that it will only process new data added to the S3 bucket since the last job run.

upvoted 2 times

linux_admin 1 year, 8 months ago

Selected Answer: A

Job bookmarks enable AWS Glue to track the data that has been processed in a previous run of the job. With job bookmarks enabled, AWS Glue will only process new data that has been added to the S3 bucket since the previous run of the job, rather than reprocessing all data every time the job runs.

upvoted 2 times

gustavtd 1 year, 11 months ago

Delete files in S3 freely is not good. so B is not correct,

upvoted 1 times

techhb 1 year, 11 months ago

Selected Answer: A

A is correct